



US012410054B2

(12) **United States Patent**
Mortensen et al.

(10) **Patent No.:** **US 12,410,054 B2**

(45) **Date of Patent:** ***Sep. 9, 2025**

(54) **SYNTHESIS GAS ON DEMAND**

(71) Applicant: **HALDOR TOPSØE A/S**, Kgs. Lyngby
(DK)

(72) Inventors: **Peter Mølgaard Mortensen**, Roskilde
(DK); **Kasper Emil Larsen**, Humlebæk
(DK); **Kim Aasberg-Petersen**, Allerød
(DK)

(73) Assignee: **HALDOR TOPSØE A/S**, Kgs. Lyngby
(DK)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 774 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **17/637,539**

(22) PCT Filed: **Sep. 24, 2020**

(86) PCT No.: **PCT/EP2020/076695**

§ 371 (c)(1),

(2) Date: **Feb. 23, 2022**

(87) PCT Pub. No.: **WO2021/063792**

PCT Pub. Date: **Apr. 8, 2021**

(65) **Prior Publication Data**

US 2022/0363537 A1 Nov. 17, 2022

(30) **Foreign Application Priority Data**

Oct. 1, 2019 (DK) PA 2019 01145

(51) **Int. Cl.**

C01B 3/38 (2006.01)

B01J 12/00 (2006.01)

(Continued)

(52) **U.S. Cl.**

CPC **C01B 3/384** (2013.01); **B01J 12/007**

(2013.01); **B01J 19/0013** (2013.01); **B01J**

19/24 (2013.01);

(Continued)

(58) **Field of Classification Search**

None

See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

3,499,947 A 3/1970 Johnson

3,871,993 A 3/1975 Morrison

(Continued)

FOREIGN PATENT DOCUMENTS

AU 2019275850 A1 11/2020

CA 2427464 A1 4/1999

(Continued)

OTHER PUBLICATIONS

U.S. Appl. No. 17/776,142, Peter Mølgaard Mortensen, filed May
11, 2022.

(Continued)

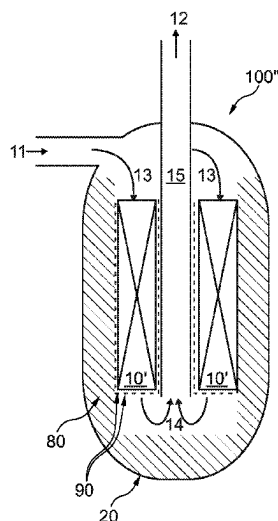
Primary Examiner — John J Figueroa

(74) *Attorney, Agent, or Firm* — Boone IP Law

(57) **ABSTRACT**

A method is provided for rapidly switching a metal-cata-
lysed steam methane reforming reaction of a feed gas from
a first steady-state reaction condition (A) to a second steady-
state reaction condition (B) or vice-versa. After applying a

(Continued)



given voltage and/or feed gas flow, the system can work towards a thermal equilibration to reach steady state without any additional operator input.

12 Claims, 11 Drawing Sheets

- (51) **Int. Cl.**
B01J 19/00 (2006.01)
B01J 19/24 (2006.01)
- (52) **U.S. Cl.**
 CPC *B01J 2219/00135* (2013.01); *C01B 2203/0233* (2013.01); *C01B 2203/085* (2013.01); *C01B 2203/1241* (2013.01)

(56) References Cited

U.S. PATENT DOCUMENTS

4,157,356	A	6/1979	Bulford et al.
4,520,216	A	5/1985	Skov et al.
4,520,224	A	5/1985	Kamimura
5,202,547	A	4/1993	Abe et al.
5,296,198	A	3/1994	Abe et al.
5,321,191	A	6/1994	Alagy et al.
5,631,302	A	5/1997	Koenig et al.
5,817,286	A	10/1998	Martin et al.
5,827,901	A	10/1998	Koenig et al.
5,976,723	A	11/1999	Boffito et al.
6,207,042	B1	3/2001	Holtermann et al.
6,322,757	B1	11/2001	Cohn et al.
6,433,029	B1	8/2002	Fitzpatrick
6,746,650	B1	6/2004	Lesieur
7,960,441	B2	6/2011	Wolf
8,568,581	B2	10/2013	Sivasankar et al.
9,067,847	B2	6/2015	Bashir et al.
9,101,890	B2 *	8/2015	Tonkovich B01F 25/431971
9,752,080	B2	9/2017	Christensen et al.
11,214,488	B2	1/2022	Rueger
12,246,298	B2 *	3/2025	Mortensen B63B 35/44
2002/0051741	A1	5/2002	Abe et al.
2002/0081253	A1	6/2002	Abe
2002/0094312	A1	7/2002	Hanus et al.
2002/0119084	A1	8/2002	Boneberg
2003/0143135	A1	7/2003	O'Rear et al.
2003/0191199	A1	10/2003	O'Rear
2004/0016650	A1	1/2004	Klug
2004/0081875	A1	4/2004	Milliken et al.
2004/0197246	A1	10/2004	Stevens et al.
2004/0265225	A1	12/2004	Watson et al.
2005/0069485	A1	3/2005	Jung et al.
2006/0116543	A1	6/2006	Bellet et al.
2006/0124445	A1	6/2006	Labrecque et al.
2006/0254141	A1	11/2006	Krause et al.
2007/0045125	A1	3/2007	Hartvigsen et al.
2008/0023338	A1	1/2008	Stoots et al.
2008/0169449	A1	7/2008	Mundschau
2009/0220390	A1	9/2009	Grouset
2009/0235587	A1	9/2009	Hawkes et al.
2009/0252919	A1	10/2009	Ogura
2009/0289227	A1	11/2009	Rising
2009/0307975	A1	12/2009	Wolf
2010/0111781	A1	5/2010	Takahashi et al.
2010/0296984	A1	11/2010	Ando et al.
2011/0020207	A1	1/2011	Siegert
2011/0136027	A1	6/2011	Chen et al.
2011/0253550	A1	10/2011	Hoffmann
2011/0253551	A1	10/2011	Lane et al.
2011/0293510	A1	12/2011	Grannell et al.
2012/0095268	A1 *	4/2012	Tonkovich B23K 31/02 422/198
2012/0201717	A1	8/2012	Singh et al.
2012/0228150	A1	9/2012	Kang et al.
2012/0288776	A1	11/2012	Nagaosa
2012/0326090	A1	12/2012	Han et al.

2013/0065974	A1	3/2013	Kresnyak
2013/0082211	A1	4/2013	Aasberg-Petersen et al.
2013/0345326	A1	12/2013	Bashir et al.
2014/0272734	A1	9/2014	Braun et al.
2014/0291162	A1	10/2014	Sala et al.
2014/0326640	A1	11/2014	De Klerk
2015/0129805	A1	5/2015	Karpenko et al.
2015/0175509	A1	6/2015	Almqvist et al.
2015/0259202	A1	9/2015	Dybkjr et al.
2015/0299871	A1	10/2015	Chen et al.
2016/0002036	A1	1/2016	Kolaczowski et al.
2016/0045841	A1 *	2/2016	Kaplan C01B 13/02 429/49
2016/0355932	A1	12/2016	Reytier et al.
2017/0106360	A1	4/2017	Meriam
2018/0066371	A1	3/2018	Hong et al.
2018/0127668	A1	5/2018	Masel
2018/0194632	A1	7/2018	Jakobsson et al.
2019/0085250	A1	3/2019	Anzelmo et al.
2019/0112187	A1	4/2019	Østberg et al.
2019/0144376	A1	5/2019	Højlund et al.
2019/0211269	A1	7/2019	Galloway
2019/0225489	A1	7/2019	Herskowitz et al.
2019/0249094	A1	8/2019	Snell et al.
2019/0359894	A1	11/2019	Heidel et al.
2020/0011225	A1	1/2020	Hirth et al.
2020/0063273	A1	2/2020	Masel
2020/0095124	A1	3/2020	Rueger
2020/0317514	A1	10/2020	Mortensen et al.
2020/0354216	A1	11/2020	Mortensen
2021/0113983	A1	4/2021	Mortensen et al.
2021/0171344	A1	6/2021	Mortensen et al.
2021/0238035	A1	8/2021	Mortensen et al.
2022/0081289	A1	3/2022	De Sarkar et al.
2022/0081291	A1 *	3/2022	Mortensen C01B 3/382
2022/0081292	A1	3/2022	De Sarkar et al.
2022/0119255	A1 *	4/2022	Mortensen B01J 8/067
2022/0162067	A1 *	5/2022	Mortensen B01J 19/2445

FOREIGN PATENT DOCUMENTS

CN	1031391	C	3/1996
CN	1483133	A	3/2004
CN	1774291	A	5/2006
CN	101157563	A	4/2008
CN	101177239	A	5/2008
CN	104169210	A	11/2014
CN	105188903	A	12/2015
CN	109964011	A	7/2019
CN	111247091	A	6/2020
CN	112203757	A	1/2021
CN	112236223	A	1/2021
DE	10104601	A1	8/2002
DE	102005046746	A1	4/2007
DE	102013102969	A1	9/2014
DE	102013226126	A1	6/2015
DE	102015110120	A1	12/2016
EP	0025205	A1	3/1981
EP	0502731	A1	9/1992
EP	0913357	A1	5/1999
EP	2491998	A1	8/2012
EP	2955158	A1	12/2015
EP	3249027	A1	11/2017
EP	2874738	B1	9/2018
EP	3415661	A1	12/2018
EP	3472370	A1	4/2019
EP	3574991	A1	12/2019
EP	3670443	A1	6/2020
GB	0722025	A	1/1955
GB	0915444	A	1/1963
GB	1269311	A	4/1972
GB	1338352	A	11/1973
GB	2358148	A	7/2001
JP	39-028629	B	12/1964
JP	S5218485	A	2/1977
JP	H5-6120	U	1/1993
JP	05-222379	A	8/1993
JP	H11130405		5/1999
JP	2002201002	A	7/2002

(56)

References Cited

FOREIGN PATENT DOCUMENTS

JP	2003-504485	A	2/2003
JP	2003226657	A	8/2003
JP	2003320254	A	11/2003
JP	2008001584	A	1/2008
JP	2010195642	A	9/2010
JP	2014-152219	A	8/2014
JP	2019-514825	A	6/2019
JP	2019-514901	A	6/2019
JP	2019-162612	A	9/2019
JP	2021-525697	A	9/2021
KR	10-2005-0030492	A	3/2005
KR	10-2009-0068427	A	6/2009
KR	10-2018-0075285	A	7/2018
KR	1020190080354		7/2019
TW	512132	B	12/2002
WO	0076651	A1	12/2000
WO	2004091773	A1	10/2004
WO	2007/019801	A1	2/2007
WO	2007/048641	A2	5/2007
WO	2007/088923	A1	8/2007
WO	2007/108014	A1	9/2007
WO	2010/004300	A1	1/2010
WO	2012/084609	A1	6/2012
WO	2013/131778	A2	9/2013
WO	2014/056535	A1	4/2014
WO	2014/096226	A1	6/2014
WO	2014/099567	A1	6/2014
WO	2014/154253	A1	10/2014
WO	2014/180888	A1	11/2014
WO	2015/014527	A1	2/2015
WO	2015/015433	A1	2/2015
WO	2016/091636	A1	6/2016
WO	2017/014635	A1	1/2017
WO	2017/036794	A1	3/2017
WO	2017/186612	A1	11/2017
WO	2017/186615	A1	11/2017
WO	2018/206235	A1	11/2018
WO	2018/228723	A1	12/2018
WO	2019/104375	A1	6/2019
WO	2019/110265	A1	6/2019
WO	2019/110267	A1	6/2019
WO	2019/110269	A1	6/2019
WO	2019110266	A1	6/2019
WO	2019110268	A1	6/2019
WO	2019/228795	A1	12/2019
WO	2019/228796	A1	12/2019
WO	2019/228797	A1	12/2019
WO	2019/228798	A1	12/2019
WO	2020/008008	A1	1/2020
WO	2020/035574	A1	2/2020
WO	2020/207926	A1	10/2020
WO	2020/208008	A1	10/2020
WO	2021/110754	A1	6/2021
WO	2022/079098	A1	4/2022

OTHER PUBLICATIONS

Keim, W., "Synthesis Gas Feedstock for Chemicals", American Chemical Society, Jan. 1, 1987, vol. 25, No. 10, pp. 1-16. (16 pages).

Kongas, Rainer, "Review-Electrochemical CO₂ Reduction for CO Production: Comparison of Low- and High-Temperature Electrolysis Technologies", Journal of the Electrochemical Society, Feb. 14, 2020, 167:0044508. (12 pages).

Wang, Y., et al., "High temperature solid oxide H₂O/Co₂ co-electrolysis for syngas production", Fuel Processing Technology, Nov. 14, 2016, vol. 161, pp. 248-258. (12 pages).

Danish Search Report for Danish Application No. PA 2019 01145 dated Mar. 12, 2020 (7 pages).

International Search Report (PCT/ISA/210) and Written Opinion (PCT/ISA/237) mailed on Nov. 23, 2020, by the European Patent Office as the International Searching Authority for International Application No. PCT/EP2020/076695.

Zhou L et al., *Investigation of a novel porous anodic alumina plate for methane steam reforming: Hydrothermal stability, electrical heating possibility and reforming reactivity*, International Journal of Hydrogen Energy, Elsevier Science Publishers B. V., vol. 34, No. 2, Jan. 1, 2009 (Jan. 1, 2009), pp. 844-858.

Danish Search Report issued in corresponding Patent Application No. PA 2019 01324 dated May 27, 2020, 8 pages.

International Search Report (PCT/ISA/210) and Written Opinion (PCT/ISA/237) mailed on Feb. 23, 2021, by the European Patent Office as the International Searching Authority for International Application No. PCT/EP2020/081700. (20 Pages).

Aasberg-Petersen, K., et al., "Synthesis gas production for FT synthesis," Studies in Surface Science and Catalysis, vol. 152, Chapter 4, 2004, p. 258-405, Elsevier B.V., the Netherlands.

Boccuzzi et al., "FTIR study of methanol decomposition on gold catalyst for fuel Cells", Journal of Power Sources, vol. 118, No. 1-2, May 25, 2003, pp. 304-310.

Danish Search Report dated Mar. 27, 2020 issued by the Danish Patent and Trademark Office in Danish Patent Application No. PA 201901437. (9 pages).

Danish Search Report for Danish Application No. PA 2019 01432 dated Mar. 27, 2020 (10 pages).

Danish Search Report issued in corresponding Patent Application No. PA 2019 01434 dated May 27, 2020 (8 pages).

European Search Report dated Jul. 11, 2018, by the European Patent Office for European Application No. 18175366.6 (7 pages).

International Preliminary Report on Patentability received for PCT Patent Application No. PCT/EP2019/062423, mailed on Dec. 10, 2020, 8 pages.

International Preliminary Report on Patentability received for PCT Patent Application No. PCT/EP2019/062424, mailed on Dec. 10, 2020, 8 pages.

International Search Report and Written Opinion received for PCT Patent Application No. PCT/EP2019/062423, mailed on Aug. 26, 2019, 10 pages.

International Search Report and Written Opinion received for PCT Patent Application No. PCT/EP2019/062424, mailed on Jul. 26, 2019, 11 pages.

International Search Report and Written Opinion received for PCT Patent Application No. PCT/EP2020/076698, mailed on Nov. 26, 2020, 9 pages.

International Search Report and Written Opinion received for PCT Patent Application No. PCT/EP2020/076700, mailed on Nov. 26, 2020, 11 pages.

International Search Report and Written Opinion received for PCT Patent Application No. PCT/EP2020/076704, mailed on Nov. 26, 2020, 11 pages.

International Search Report and Written Opinion received for PCT Patent Application No. PCT/EP2020/076707, mailed on Nov. 27, 2020, 11 pages.

International Search Report and Written Opinion received for PCT Patent Application No. PCT/EP2020/076713, mailed on Nov. 26, 2020, 11 pages.

Search Report dated Apr. 24, 2020, issued in the Danish Patent Application No. PA201901433, 9 pages.

Search Report dated Apr. 24, 2020, issued in the Danish Patent Application No. PA201901435, 9 pages.

Search Report dated Jan. 28, 2019, issued in the Danish Patent Application No. PA201800249, 9 pages.

Wismann, Sebastian T., et al., "Electrified methane reforming: A compact approach to greener industrial hydrogen production," Science, May 24, 2019, p. 756-759, vol. 364, American Association for the Advancement of Science, Washington, D.C.

Xu et al., "Methane Steam Reforming, Methanation and Water-Gas Shift: I. Intrinsic Kinetics", American Institution of Chemical Engineers Journal, vol. 35, No. 1, Jan. 1989, pp. 88-96.

U.S. Appl. No. 17/046,475 (Cited herein as US Patent Application Publication No. 2021/0171344 A1 of Jun. 10, 2021), Peter Mølgaard Mortensen, filed Oct. 9, 2020.

U.S. Appl. No. 17/054,572 (Cited herein as US Patent Application Publication No. 2021/0113983 A1 of Apr. 22, 2021), Peter Mølgaard Mortensen, filed Nov. 11, 2020.

(56)

References Cited**OTHER PUBLICATIONS**

U.S. Appl. No. 17/627,202, Peter Mølgaard Mortensen, filed Jan. 14, 2022.

U.S. Appl. No. 17/630,734, Peter Mølgaard Mortensen, filed Jan. 27, 2022.

U.S. Appl. No. 17/636,945, Peter Mølgaard Mortensen, filed Feb. 21, 2022.

U.S. Appl. No. 17/638,423, Peter Mølgaard Mortensen, filed Feb. 25, 2022.

U.S. Appl. No. 17/641,293, Peter Mølgaard Mortensen, filed Mar. 8, 2022.

Technology for Application of Industrial Control Computers, pp. 303-304, Beijing: Chemical Engineering Press, May 1982.

Introduction to Energy Chemistry, Dong Guanghua, etc., p. 124, Xuzhou: China University of Mining and Technology Press, Sep. 2018.

Office Action with English translation only, mailed on Jul. 18, 2024, by the China National Intellectual Property Administration for Chinese Application No. 2020800688566, 7 pages.

Bonis, L.J. and H.H. Hausner, Fundamental Phenomena in the Material Sciences, vol. 1: Sintering and Plastic Deformation, 1964, pp. 3-101, 1964.

U.S. Appl. No. 18/257,339 filed Jun. 14, 2023, Kim Aasber-Petersen.

U.S. Appl. No. 18/990,223 filed Dec. 20, 2024, Peter Mølgaard Mortensen.

Communication dated Mar. 25, 2021 and Search Report dated Mar. 17, 2021 issued in corresponding European Patent Application No. 20201822.2. (7 pages).

Decision to Grant a Patent received for Japanese Patent Application No. 2020-566667, mailed on Dec. 6, 2023, 5 pages (2 pages of English Translation and 3 pages of Original Document).

Decision to Grant a Patent received for Japanese Patent Application No. 2022-520208, mailed on Jan. 8, 2025, 05 pages (02 pages of English Translation and 03 pages of Original Document).

Decision to Grant a Patent received for Japanese Patent Application No. 2022-520209, mailed on Mar. 12, 2025, 05 pages (02 pages of English Translation and 03 pages of Original Document).

Decision to Grant a Patent received for Japanese Patent Application No. 2022-520211, mailed on Feb. 5, 2025, 05 pages (02 pages of English Translation and 03 pages of Original Document).

Decision to Grant a Patent received for Japanese Patent Application No. 2022-526352, mailed on Mar. 26, 2025, 05 pages (02 pages of English Translation and 03 pages of Original Document).

European Search Report issued in corresponding Application No. EP 19 21 3432 dated May 8, 2020.

European Search Report issued in corresponding Application No. EP 20 20 1817 dated Mar. 17, 2021.

Extended European Search Report dated Mar. 25, 2021, issued by the European Patent Office in corresponding European Application No. 20201816.4. (8 pages).

Extended European Search Report mailed on Jul. 8, 2021, by the European Patent Office for European Application No. (2115595534).

First Office Action with English translation mailed on September 3, 2023, by the Saudi Authority for Intellectual Property for Saudi Arabian Application No. 522432548, 19 pages.

International Preliminary Report on Patentability received for PCT Patent Application No. PCT/EP2020/084408, mailed on Jun. 16, 2022, 8 pages.

International Preliminary Report on Patentability received for PCT Patent Application No. PCT/EP2021/078142, mailed on Apr. 27, 2023, 8 pages.

International Preliminary Report on Patentability received for PCT Patent Application No. PCT/EP2021/078304, mailed on Apr. 27, 2023, 8 pages.

International Preliminary Report on Patentability received for PCT Patent Application No. PCT/EP2022/051090, mailed on Aug. 10, 2023, 11 pages.

International Preliminary Report on Patentability received for PCT Patent Application No. PCT/EP2022/053062, mailed on Aug. 24, 2023, 11 pages.

International Search Report and Written Opinion received for PCT Patent Application No. PCT/EP2020/084408, mailed on Feb. 12, 2021, 11 pages.

International Search Report and Written Opinion received for PCT Patent Application No. PCT/EP2021/078142, mailed on Jan. 4, 2022, 11 pages.

International Search Report and Written Opinion received for PCT Patent Application No. PCT/EP2022/051090, mailed on May 10, 2022, 14 pages.

International Search Report and Written Opinion received for PCT Patent Application No. PCT/EP2022/053062, mailed on May 12, 2022, 13 pages.

International Search Report and Written Opinion received for PCT Patent Application No. PCT/EP21/078304, mailed on Dec. 23, 2021, 11 pages.

Notice of Allowance received for Chinese Patent Application No. 201980036811, mailed on May 2, 2025, 4 (2 pages of English Translation and 2 pages of Original Document).

Notification to grant right for invention received for Chinese Patent Application No. 202080067396.5, mailed on Sep. 26, 2024, 03 pages (02 pages of English Translation and 01 pages of Original Document).

Office Action received for Chinese Patent Application No. 201980034543, mailed on Jan. 24, 2022, 17 pages (9 pages of English Translation and 8 pages of Original Document).

Office Action received for Chinese Patent Application No. 201980036811, mailed on Feb. 9, 2023, 7 pages (4 pages of English Translation and 3 pages of Original Document).

Office Action received for Chinese Patent Application No. 202080067342, mailed on Feb. 15, 2023, 19 pages (9 pages of English Translation and 10 pages of Original Document).

Office Action received for Chinese Patent Application No. 202080067342.9, mailed on Jan. 4, 2024, 6 pages.

Office Action received for Chinese Patent Application No. 202080067342.9, mailed on Sep. 20, 2023, 17 pages. (10 pages of English Translation and 7 pages of Original Document).

Office Action received for Chinese Patent Application No. 202080067396.5, mailed on Mar. 7, 2024, 22 pages. (14 pages of English Translation and 8 pages of Original Document).

Office Action received for Chinese Patent Application No. 202080068829.9, mailed on Jul. 17, 2024, 08 pages.

Office Action received for Chinese Patent Application No. 202080068829.9, mailed on Apr. 7, 2023, 18 pages (12 pages of English Translation and 6 pages of Original Document).

Office Action received for Chinese Patent Application No. 202080068829.9, mailed on Apr. 22, 2024, 18 pages. (11 pages of English Translation and 7 pages of Original Document).

Office Action received for Chinese Patent Application No. 202080068829.9, mailed on Nov. 10, 2023, 17 pages (10 pages of English Translation and 7 pages of Original Document).

Office Action received for Chinese Patent Application No. 202080068856.6, mailed on Nov. 9, 2023, 15 pages. (9 pages of English Translation and 6 pages of Original Document).

Office Action received for Chinese Patent Application No. 202080068856.6, mailed on Apr. 6, 2023, 16 pages. (11 pages of English Translation and 5 pages of Original Document).

Office Action received for Chinese Patent Application No. 202080068856.6, mailed on Apr. 23, 2024, 16 pages. (10 pages of English Translation and 6 pages of Original Document).

Office Action received for Chinese Patent Application No. 202080069164.3, mailed on Apr. 6, 2023, 18 pages. (12 pages of English Translation and 6 pages of Original Document).

Office Action received for Chinese Patent Application No. 202080069164.3, mailed on Jan. 4, 2024, 6 pages.

Office Action received for Chinese Patent Application No. 202080069164.3, mailed on Nov. 9, 2023, 18 pages. (11 pages of English Translation and 7 pages of Original Document).

Office Action received for Chinese Patent Application No. 202080070202.7, mailed on Sep. 20, 2023, 20 pages. (13 pages of English Translation and 7 pages of Original Document).

(56)

References Cited

OTHER PUBLICATIONS

Office Action received for Chinese Patent Application No. 202080078792.8, mailed on Mar. 25, 2023, 29 pages. (18 pages of English Translation and 11 pages of Original Document).

Office Action received for Chinese Patent Application No. 202080078792.8, mailed on Mar. 30, 2024, 16 pages.

Office Action received for Chinese Patent Application No. 202080078792.8, mailed on Oct. 12, 2023, 25 pages. (16 pages of English Translation and 9 pages of Original Document).

Office Action received for European Application No. 20816477, mailed on Aug. 16, 2023, 6 pages.

Office Action received for European Patent Application No. 20780175, mailed on Apr. 22, 2025, 3 pages.

Office Action received for Japanese Patent Application No. 2022-520208, mailed on Aug. 7, 2024, 07 pages (03 pages of English Translation and 04 pages of Original Document).

Office Action received for Japanese Patent Application No. 2022-520209, mailed on Oct. 11, 2024, 10 pages (05 pages of English Translation and 05 pages of Original Document).

Office Action received for Japanese Patent Application No. 2022-520210, mailed on Oct. 11, 2024, 09 pages (04 pages of English Translation and 05 pages of Original Document).

Office Action received for Japanese Patent Application No. 2022-520211, mailed on Sep. 30, 2024, 06 pages (03 pages of English Translation and 03 pages of Original Document).

Office Action received for Japanese Patent Application No. 2022-526352, mailed on Dec. 11, 2024, 06 pages (03 pages of English Translation and 03 pages of Original Document).

Request for the Submission of an Opinion received for Korean Patent Application No. 10-2022-7012146, mailed on Mar. 4, 2025, 08 pages (4 pages of English Translation and 4 pages of Original Document).

Request for the Submission of an Opinion received for Korean Patent Application No. 10-2020-7037585, mailed on May 1, 2024, 08 pages (4 pages of English Translation and 4 pages of Original Document).

Request for the Submission of an Opinion received for Korean Patent Application No. 10-2020-7037588, mailed on May 1, 2024, 8 pages (4 pages of English Translation and 4 pages of Original Document).

Written Decision on Registration received for Korean Patent Application No. 10-2020-7037585, mailed on Feb. 19, 2025, 5 pages (3 pages of English Translation and 2 pages of Original Document).

Written Decision on Registration received for Korean Patent Application No. 10-2020-7037588, mailed on Feb. 19, 2025, 5 pages (3 pages of English Translation and 2 pages of Original Document).

Wu et al., "Methanation of CO₂ and reverse water gas shift reaction on Ni/SiO₂ catalysts: the influence of particle size on selectivity and reaction pathway", Catalysis Science & Technology, vol. 5, No. 8, Jan. 1, 2015, pp. 4154-4163.

Yabe, Tomohiro et al., "Low-temperature dry reforming of methane to produce syngas in an electric field over La-doped Ni/ZrO₂ catalysts", Fuel Processing Technology, Dec. 23, 2016, vol. 158, pp. 96-103, Elsevier B.V., NL. (8 pages).

* cited by examiner

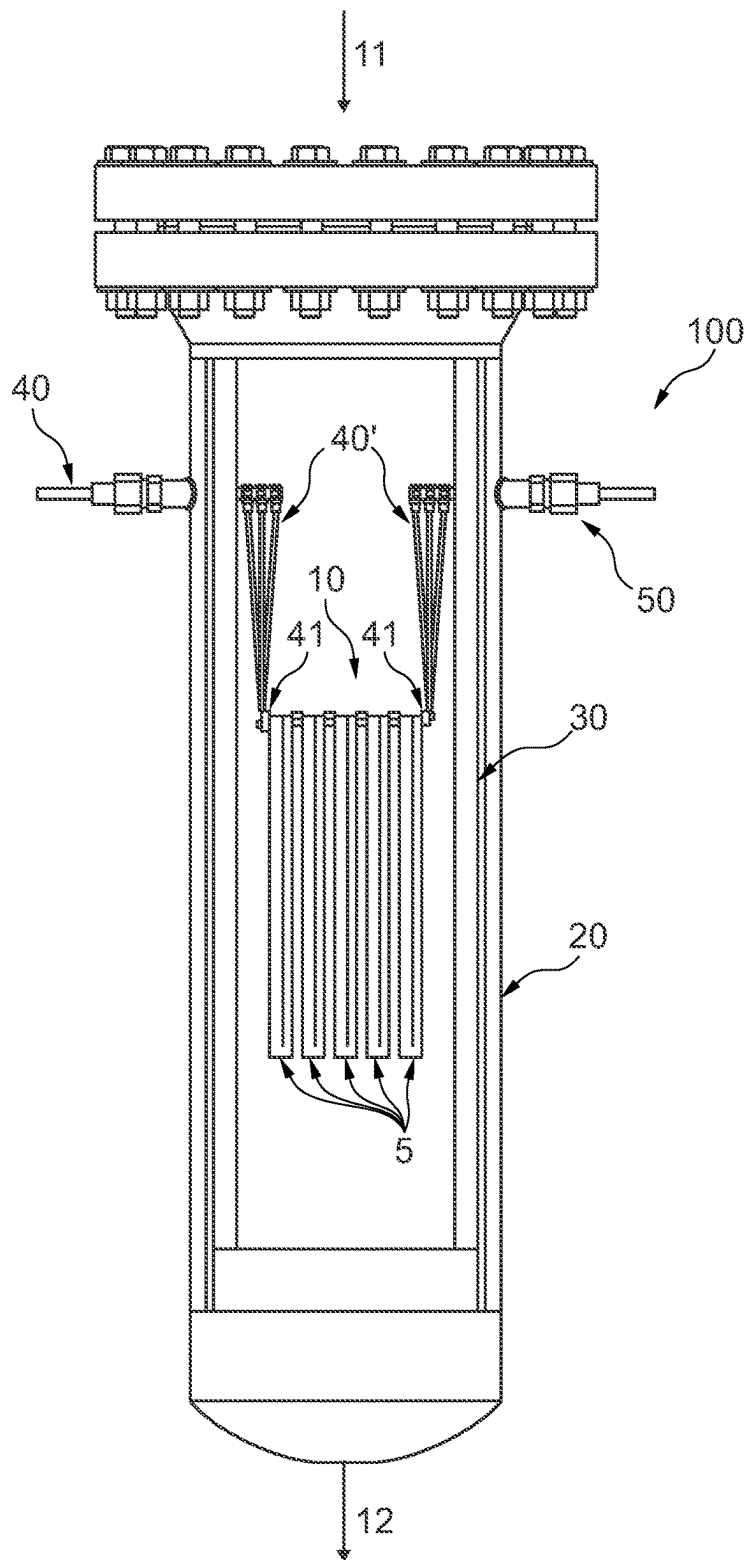


Fig. 1a

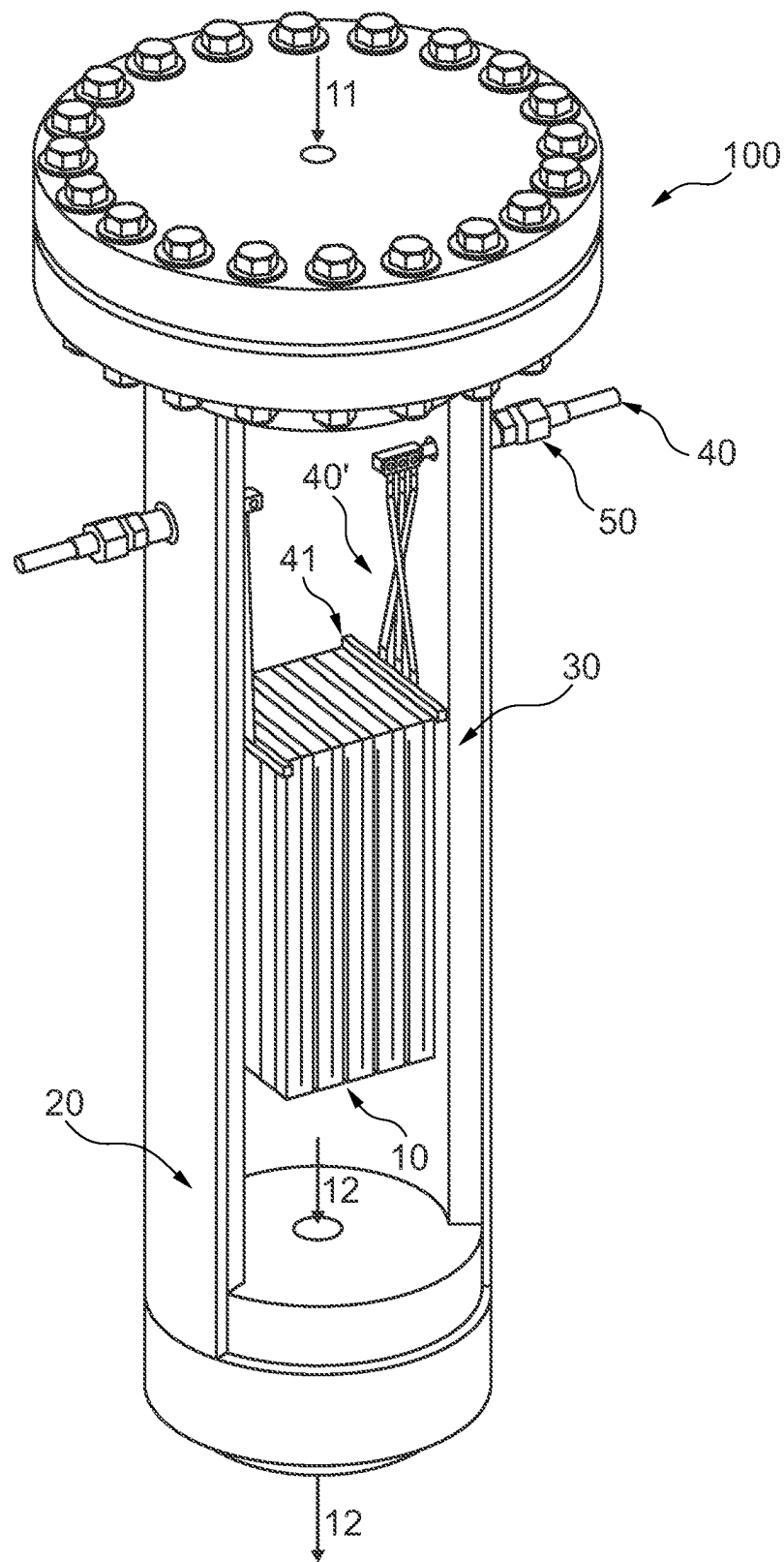


Fig. 1b

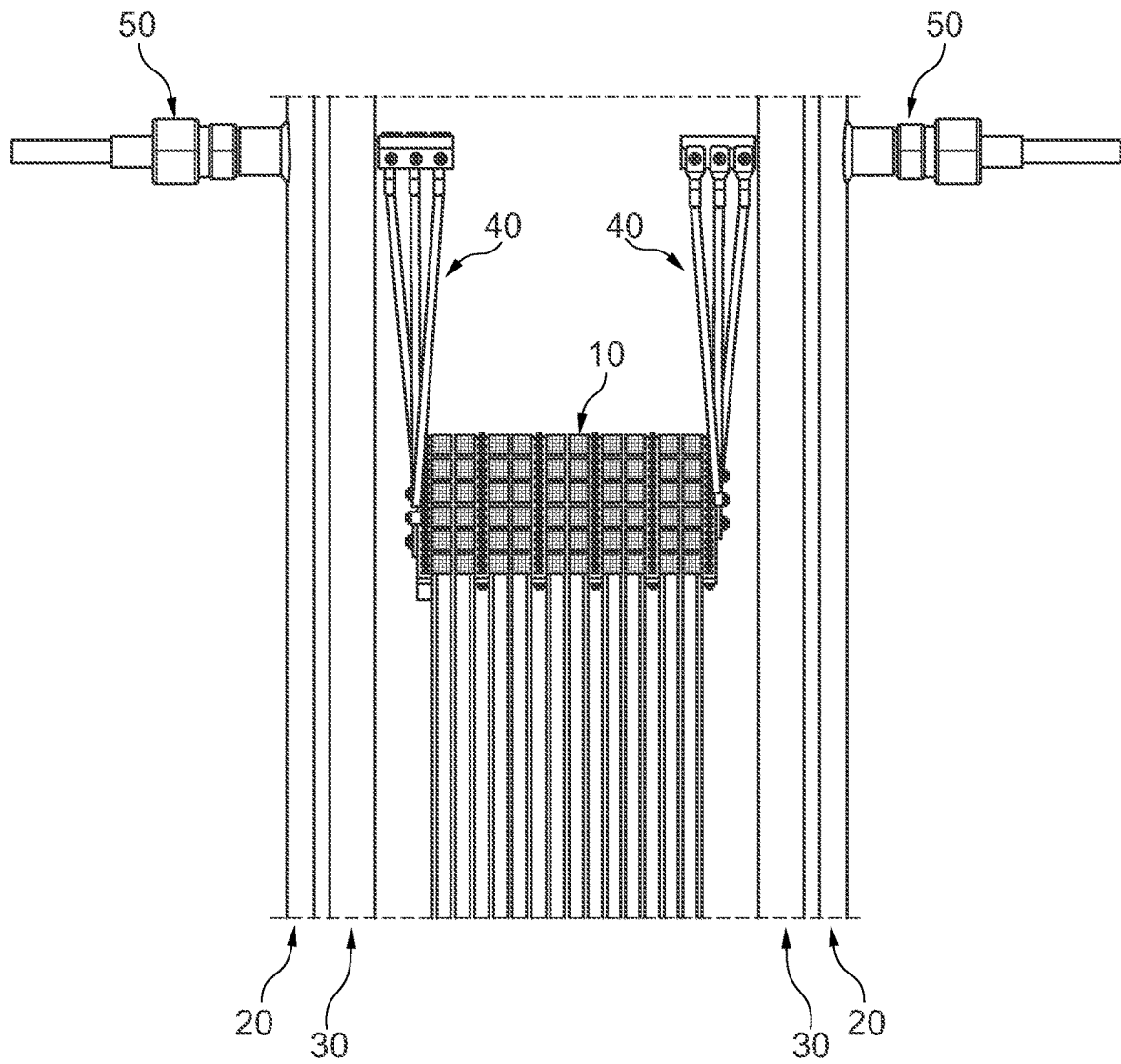


Fig. 2

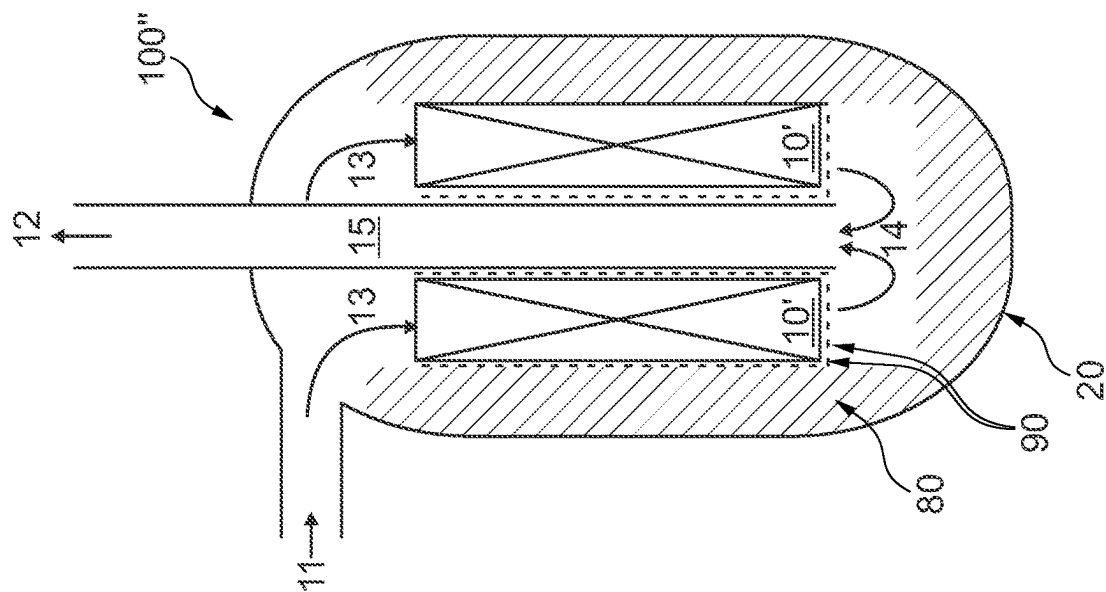


Fig. 3a

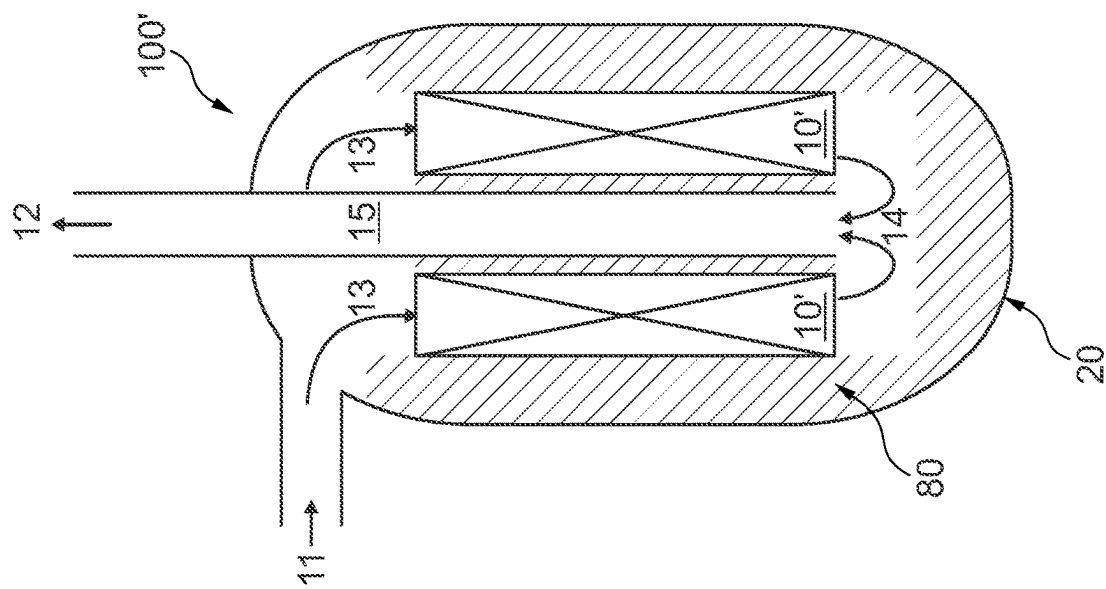


Fig. 3b

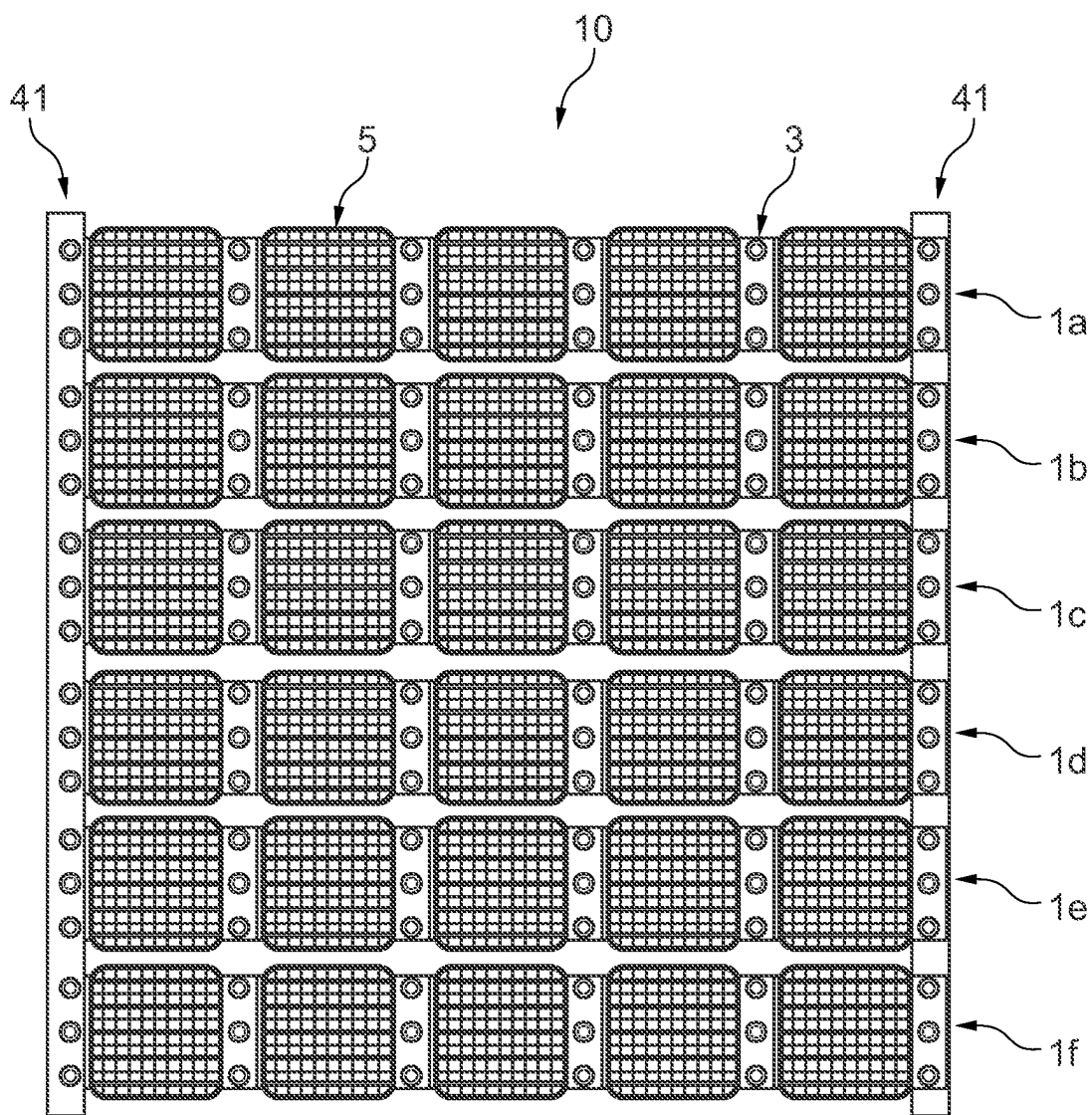


Fig. 4

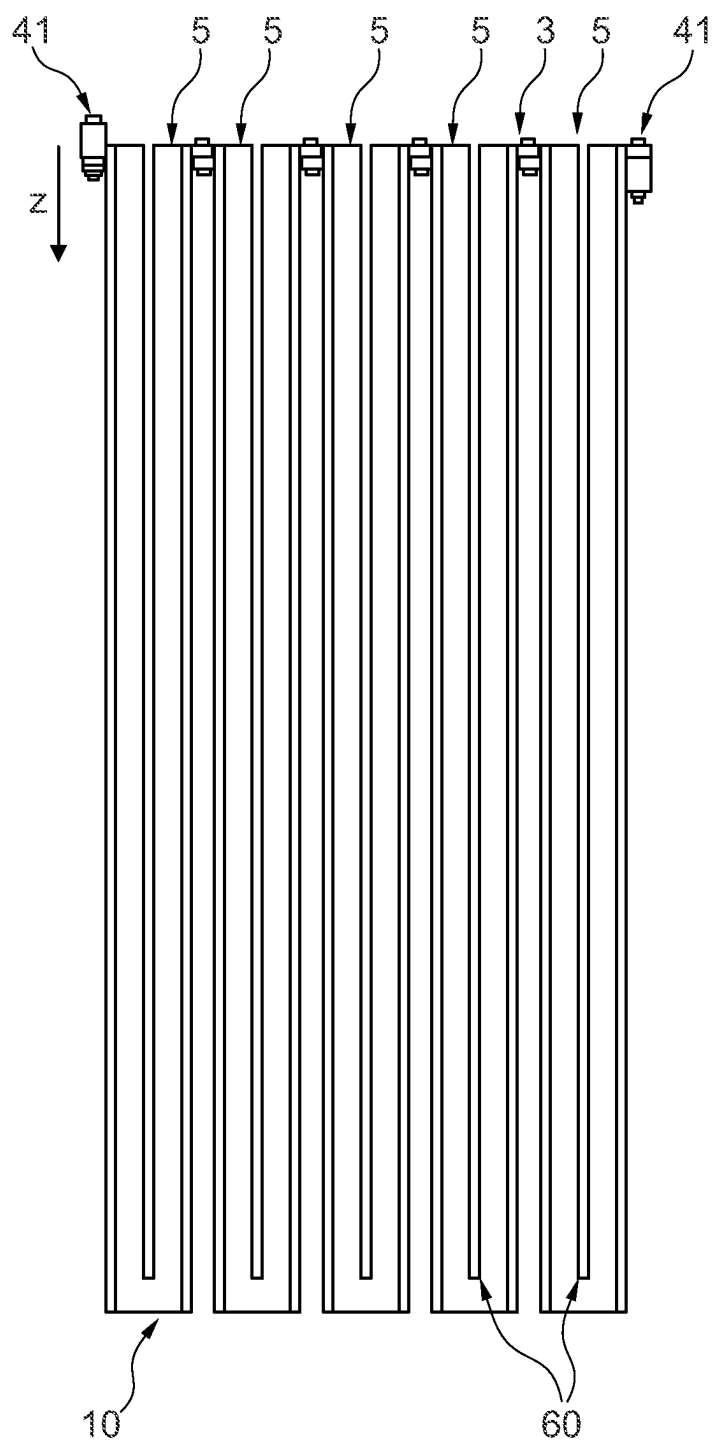


Fig. 5

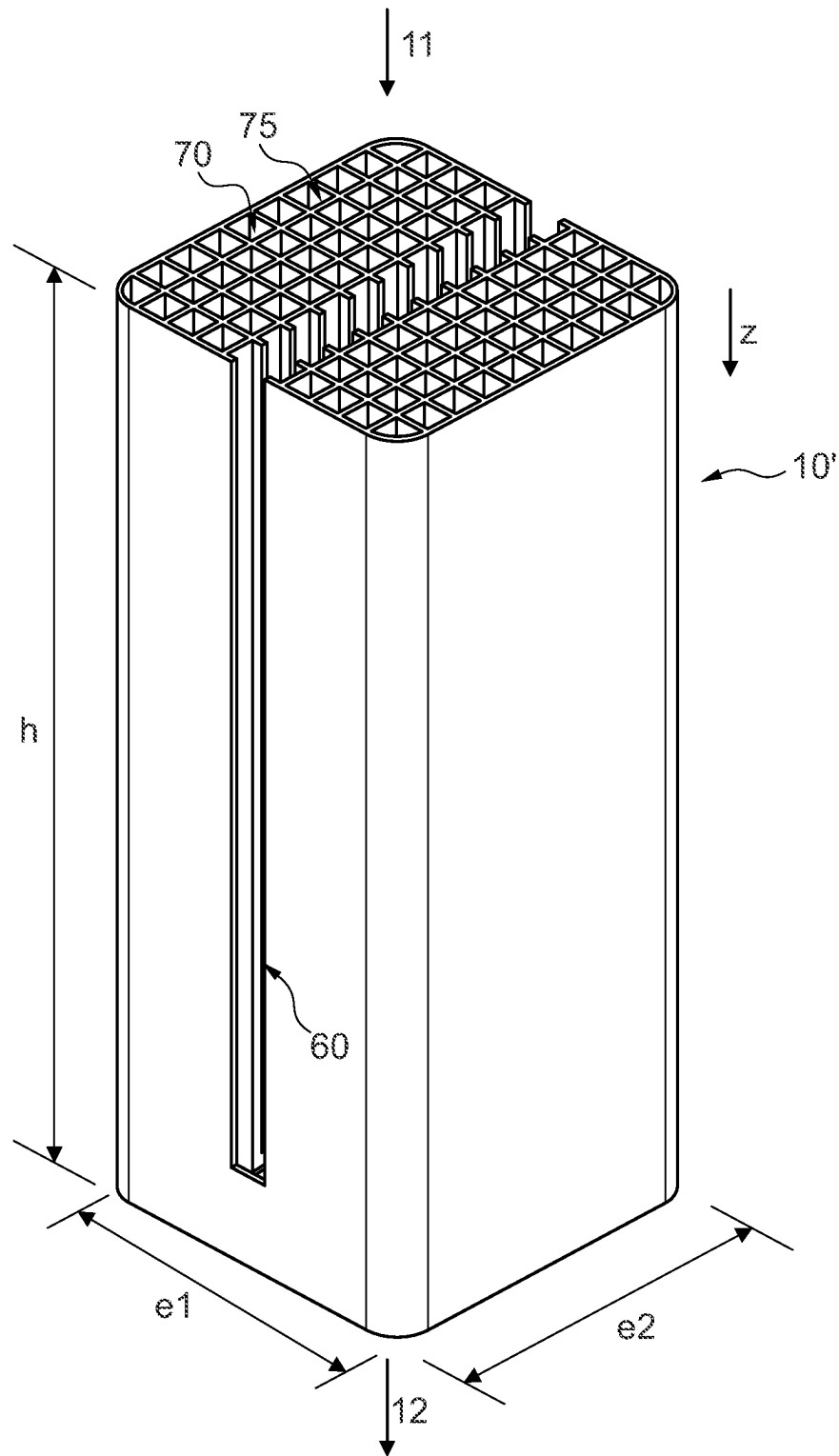


Fig. 6

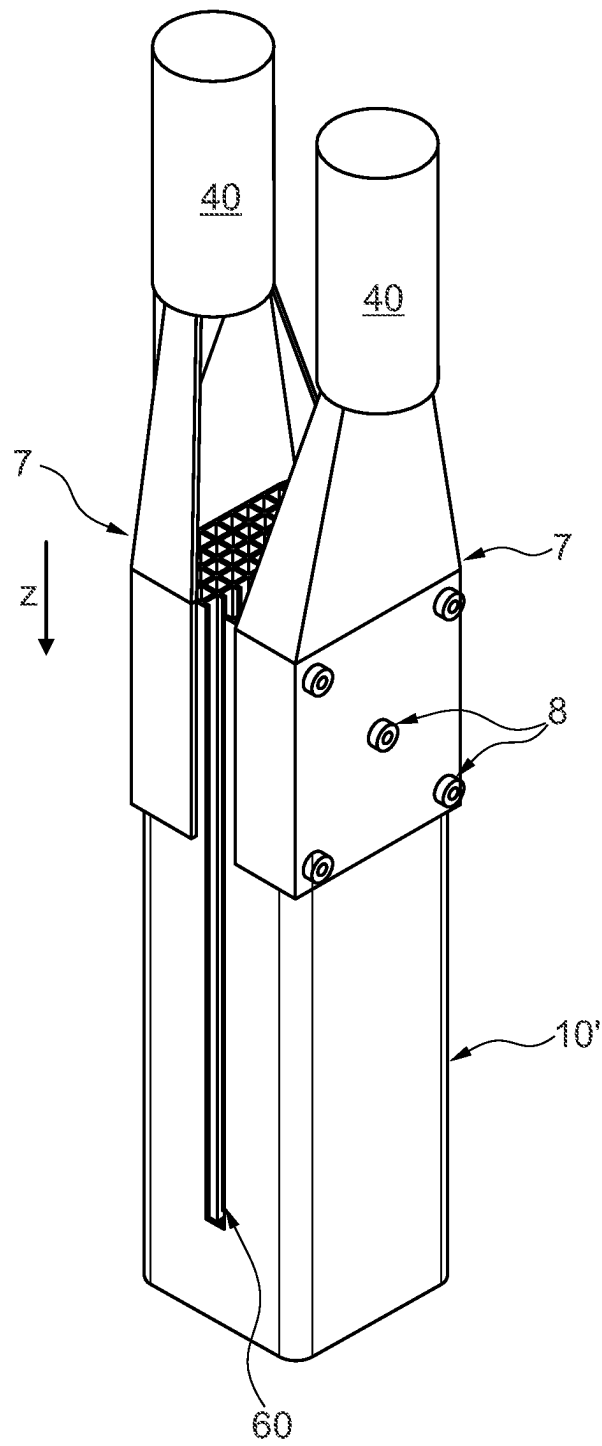


Fig. 7

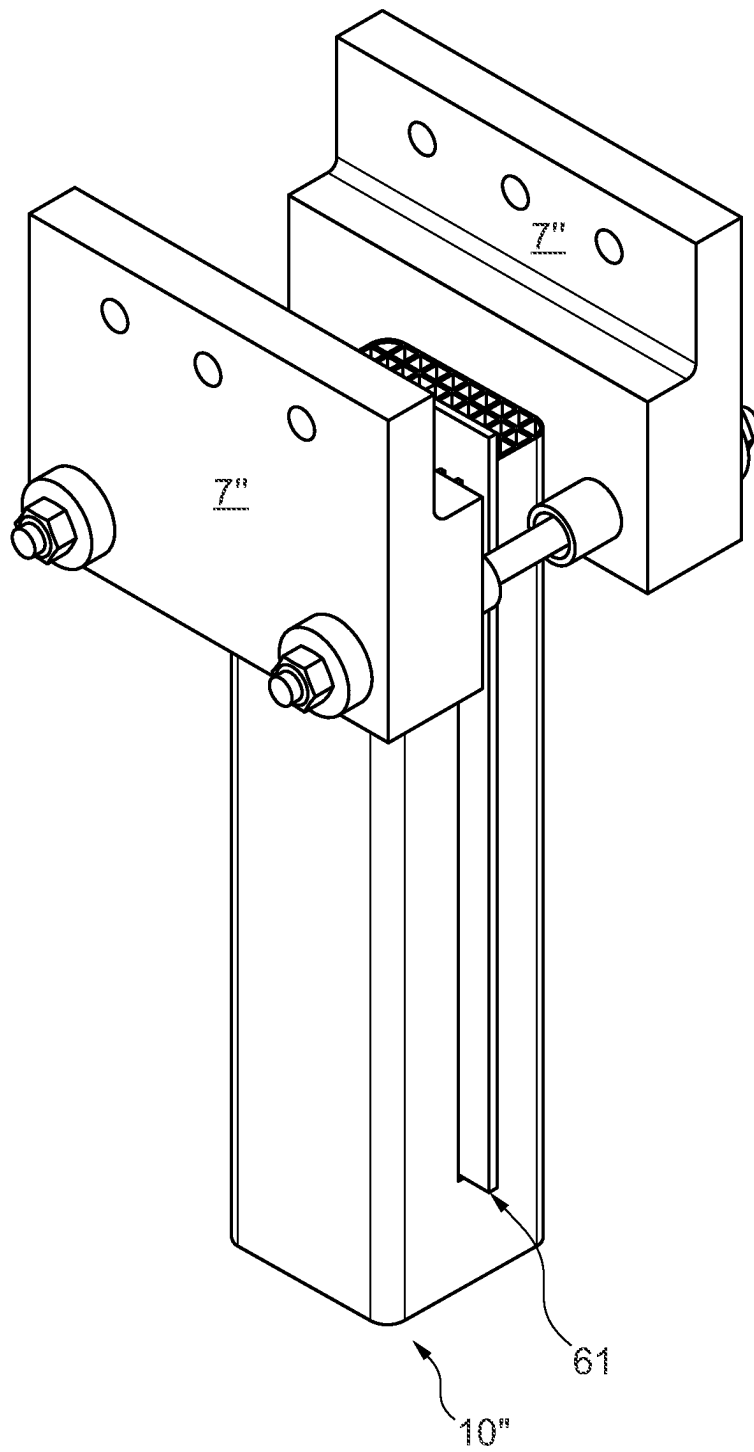


Fig. 8

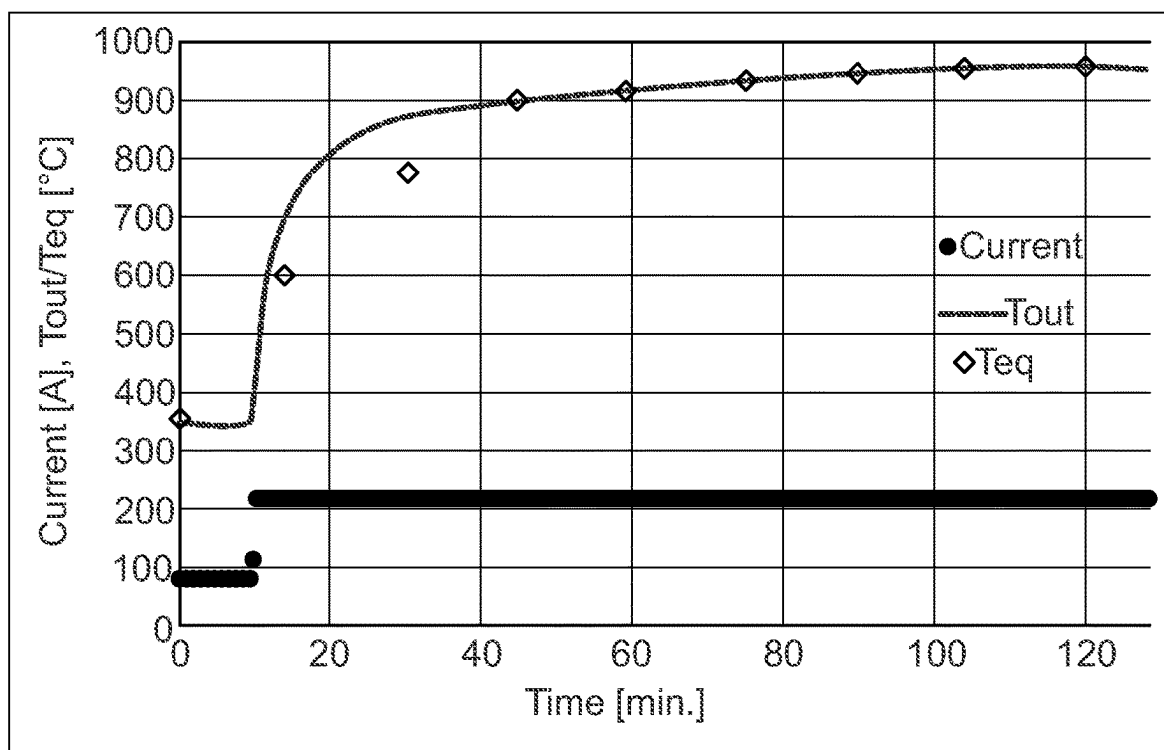


Fig. 9

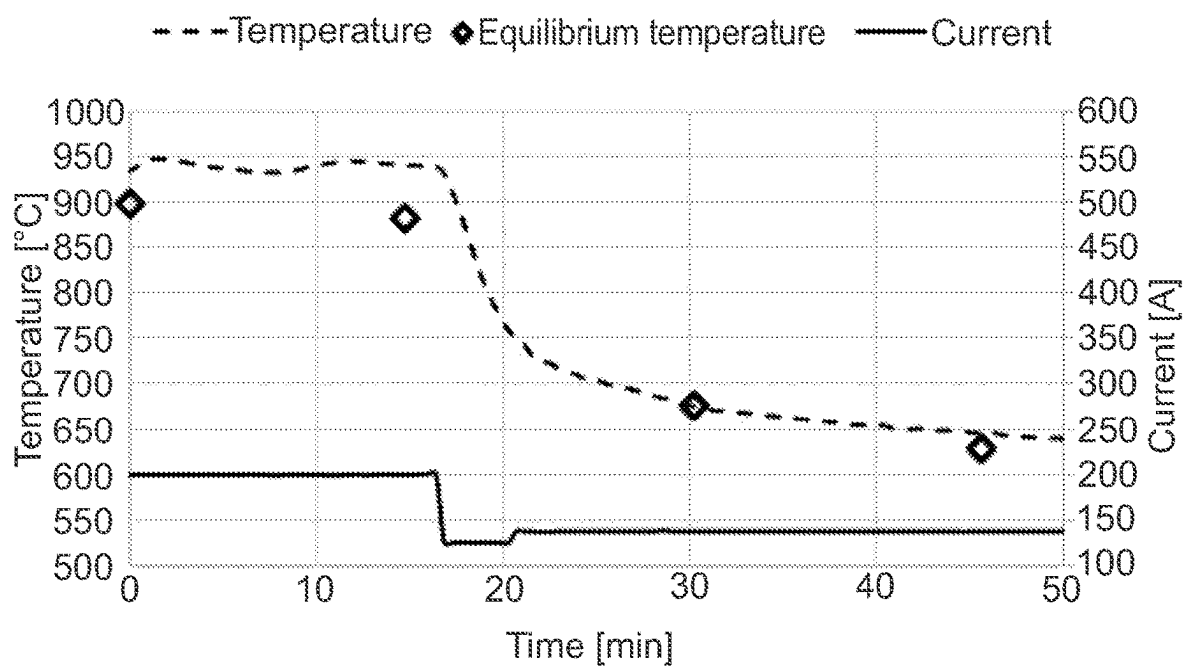


Fig. 10

1

SYNTHESIS GAS ON DEMAND

TECHNICAL FIELD

A method is provided for rapidly switching a metal-catalysed steam methane reforming reaction of a feed gas from a first steady-state reaction condition (A) to a second steady-state reaction condition (B) or vice-versa.

BACKGROUND

Steam methane reforming (SMR) reactors are typically difficult to start-up due to the requirements of higher temperature chemistry and the associated heating. Taking the SMR reactor as an example, the fired reformer technology works by providing the energy input by combustion externally to the catalytic reactor system. In practice, this means that the process must be controlled by balancing two chemical reactors against each other; on one side the combustion reactor and on the other side the catalytic reactor. At all time, these must be balanced, and—in particular—the combustion side is not allowed a large offset from the catalyst side, otherwise overheating and thereby mechanical failure will occur.

It only adds increased complexity that the catalytic reactor side of a typical SMR reactor is configured as several parallel tubes with one tube being able to process in the order of 100-300 Nm³/h gas and added capacity is gained by multiplying the number of tubes. In practice, this means that fired reformers are tedious to operate and care must be taken to maintain the process in steady state where an even degree of heat and gas distribution between the individual tubes is obtained.

Consequently, the ease by which steam methane reforming unit changes from one steady-state condition to another is typically slow because heating needs to be balanced against heat consumption. ATR technology also suffers from difficult start-up, where preheating is essential to avoid excessive sooting from the flame in the reactor. In some instances this requires preheating of large amounts of non-reactive gas before start-up is possible. This also influences the size and complexity of utility systems for such a plant.

Related technology is disclosed in applications PCT/EP2019/062423 and PCT/EP2019/062424.

One aim of the present technology is therefore to reduce the time it takes for a steam reforming reactor to change from one steady-state reaction condition to another, e.g. during start-up. Another aim of the present technology is to improve and/or simplify control of a steam reforming reaction in a steam reforming unit.

SUMMARY

A method for rapidly switching a metal-catalysed steam methane reforming reaction of a feed gas comprising hydrocarbons in a reactor system from a first steady-state reaction condition (A) to a second steady-state reaction condition (B) or vice-versa is provided.

The reactor system comprises a pressure shell housing a structured catalyst arranged to catalyze steam reforming of a feed gas comprising hydrocarbons, said structured catalyst comprising a macroscopic structure of an electrically conductive material, said macroscopic structure supporting a ceramic coating, where said ceramic coating supports a catalytically active material and wherein said reactor system is provided with heat insulation between said structured catalyst and said pressure shell and where a power supply

2

placed outside said pressure shell is arranged to supply electrical power via electrical conductors connecting to said structured catalyst, allowing an electrical current to run through said macroscopic structure, thereby heating at least part of the structured catalyst.

The method comprises the steps of:

in said first steady-state reaction condition (A):

supplying said feed gas to the reactor system in a first total flow, and

supplying a first electrical power via electrical conductors connecting an electrical power supply placed outside said pressure shell to said structured catalyst, thereby allowing a first electrical current to run through said electrically conductive material,

thereby heating at least part of the structured catalyst to a first temperature at which said feed gas is converted to a first product gas mixture over said structured catalyst under said first steady-state reaction conditions (A); and said first product gas is outlet from the reactor system;

and, in said second steady-state reaction condition (B):

supplying said feed gas to the reactor system in a second total flow,

supplying a second electrical power via electrical conductors connecting an electrical power supply placed outside said pressure shell to said structured catalyst, thereby allowing a second electrical current to run through said electrically conductive material,

thereby heating at least part of the structured catalyst to a second temperature; at which said feed gas is converted to a second product gas mixture over said structured catalyst under said second steady-state reaction conditions (B); and said second product gas is outlet from the reactor system;

wherein said second electrical power is higher than said first electrical power; and/or said second total flow is higher than said first total flow.

Additional aspects of the present technology are set out in the dependent claims, the figures and the following description text.

LEGENDS TO THE FIGURES

FIG. 1a shows a cross section through an embodiment of the inventive reactor system with a structured catalyst comprising an array of macroscopic structures, in a cross section;

FIG. 1b shows the reactor system of FIG. 1a with a part of the pressure shell and heat insulation layer removed;

FIG. 2 is an enlarged view of a part of the reactor system;

FIGS. 3a and 3b show schematic cross sections through an embodiment of the inventive reactor system comprising a structured catalyst;

FIGS. 4 and 5 show an embodiment of a structured catalyst with an array of macroscopic structures as seen from above and from the side, respectively;

FIG. 6 shows an embodiment of the structured catalyst of the invention;

FIGS. 7 and 8 show embodiments of a structured catalyst with connectors;

FIGS. 9 and 10 show the transient behaviour during an experiment of an embodiment of the invention.

DETAILED DISCLOSURE

A reactor system for carrying out an endothermic reaction of a feed gas, is generally provided, the reactor system comprising:

3

a structured catalyst arranged for catalyzing the endothermic reaction of a feed gas, where the structured catalyst comprises a macroscopic structure of electrically conductive material, where the macroscopic structure supports a ceramic coating and wherein the ceramic coating supports a catalytically active material;

a structured catalyst arranged for catalyzing said endothermic reaction of a feed gas, said structured catalyst comprising an electrically conductive material and a catalyst material;

a pressure shell housing said structured catalyst;

a heat insulation layer between said structured catalyst and said pressure shell;

a power supply placed outside said pressure shell is arranged to supply electrical power via electrical conductors connecting to said structured catalyst, allowing an electrical current to run through said macroscopic structure, thereby heating at least part of the structured catalyst

at least two conductors electrically connected to said structured catalyst and to an electrical power supply placed outside the pressure shell, wherein the electrical power supply is dimensioned to heat at least part of the structured catalyst to a temperature of at least 200° C. by passing an electrical current through the electrically conductive material.

The layout of the reactor system allows for feeding a pressurized feed gas to the reactor system at an inlet and directing this gas into the pressure shell of the reactor system. Inside the pressure shell, a configuration of heat insulation layers and inert material is arranged to direct the feed gas through the structured catalyst where it will be in contact with the catalyst material, where the catalytically active material will facilitate the steam reforming reaction. Additionally, the heating of the structured catalyst will supply the required heat for the endothermic reaction. The product gas from the heated structured catalyst is led to the reactor system outlet.

The close proximity between the catalytically active material and the electrically conductive materials enables efficient heating of the catalytically active material by close proximity heat conduction from the resistance heated electrically conductive material. An important feature of the resistance heating process is thus that the energy is supplied inside the object itself, instead of being supplied from an external heat source via heat conduction, convection and radiation. Moreover, the hottest part of the reactor system will be within the pressure shell of the reactor system. Preferably, the electrical power supply and the structured catalyst are dimensioned so that at least part of the structured catalyst reaches a temperature of 850-1100° C. when the endothermic reaction is the steam reforming reaction, a temperature of 700-1200° C. when the endothermic reaction is the hydrogen cyanide synthesis, a temperature of 500-700° C. when the endothermic reaction is dehydrogenation, a temperature of 200-300° C. when the endothermic reaction is the methanol cracking, and a temperature of ca. 500° C. when the endothermic reaction is the ammonia cracking reaction. The surface area of the electrically conductive material, the fraction of the electrically conductive material coated with a ceramic coating, the type and structure of the ceramic coating, and the amount and composition of the catalytically active catalyst material may be tailored to the specific endothermic reaction at the given operating conditions.

The electrically conductive material is suitably a macroscopic structure. As used herein, the term “macroscopic

4

structure” is meant to denote a structure that is large enough to be visible with the naked eye, without magnifying devices. The dimensions of the macroscopic structure are typically in the range of centimeters or even meters. Dimensions of the macroscopic structure are advantageously made to correspond at least partly to the inner dimensions of the pressure shell housing the structured catalyst, saving room for the heat insulation layer and conductors. Two or more macroscopic structures may be connected in order to provide an array of macroscopic structures having at least one of the outer dimensions in the range of meters, such as 2 m or 5 m. Such two or more macroscopic structures may be denoted “an array of macroscopic structures”. In this case the dimensions of an array of macroscopic structures are advantageously made to correspond at least partly to the inner dimension of the pressure shell housing the structured catalyst (saving room for the heat insulation layer). A conceivable array of macroscopic structures could take up a volume of 0.1 to 10 m³ or even larger. The structured catalyst may comprise a single macroscopic structure or an array of macroscopic structures, where the macroscopic structure(s) support(s) a ceramic coating supporting catalytically active material. In an array of macroscopic structures, the macroscopic structures may be electrically connected to each other; however, alternatively, the macroscopic structures are not electrically connected to each other. Thus, the structured catalyst may comprise two or more macroscopic structures positioned adjacent to each other. The macroscopic structure(s) may be extruded and sintered structures or 3D printed structures. A 3D printed macroscopic structure can be provided with or without subsequent sintering.

The physical dimensions of the macroscopic structure may be any appropriate dimensions; thus, the height may be smaller than the width of the macroscopic structure or vice versa.

The macroscopic structure supports a ceramic coating, where the ceramic coating supports a catalytically active material. The term “macroscopic structure supporting a ceramic coating” is meant to denote that the macroscopic structure is coated by the ceramic coating at, at least, a part of the surface of the macroscopic structure. Thus, the term does not imply that all the surface of the macroscopic structure is coated by the ceramic coating; in particular, at least the parts of the macroscopic structure which are electrically connected to the conductors do not have a coating thereon. The coating is a ceramic material with pores in the structure, which allows for supporting catalytically active material on and inside the coating. Advantageously, the catalytically active material comprises catalytically active particles having a size in the range from about 5 nm to about 250 nm.

Preferably, the macroscopic structure has been manufactured by extrusion of a mixture of powdered metallic particles and a binder to an extruded structure and subsequent sintering of the extruded structure, thereby providing a material with a high geometric surface area per volume. Preferably, the extruded structure is sintered in a reducing atmosphere to provide the macroscopic structure. Alternatively, the macroscopic structure is 3D printed a metal additive manufacturing melting process, viz. a 3D printing processes, which do not require subsequent sintering, such as powder bed fusion or direct energy deposition processes. Examples of such powder bed fusion or direct energy deposition processes are laser beam, electron beam or plasma 3D printing processes. As another alternative, the macroscopic structure may have been manufactured as a 3D metal structure by means of a binder-based metal additive

5

manufacturing process, and subsequent sintered in a non-oxidizing atmosphere at a first temperature T_1 , where $T_1 > 1000^\circ \text{C}$., in order to provide the macroscopic structure.

A ceramic coating, which may contain the catalytically active material, is provided onto the macroscopic structure before a second sintering in an oxidizing atmosphere, in order to form chemical bonds between the ceramic coating and the macroscopic structure. Alternatively, the catalytically active material may be impregnated onto the ceramic coating after the second sintering. When chemical bonds are formed between the ceramic coating and the macroscopic structure an especially high heat conductivity between the electrically heated macroscopic structure and the catalytically active material supported by the ceramic coating is possible, offering close and nearly direct contact between the heat source and the catalytically active material of the structured catalyst. Due to close proximity between the heat source and the catalytically active material the heat transfer is effective, so that the structured catalyst can be very efficiently heated. A compact reactor system in terms of gas processing per reactor system volume is thus possible, and therefore the reactor system housing the structured catalyst may be compact.

As used herein, the terms "3D print" and "3D printing" is meant to denote a metal additive manufacturing process. Such metal additive manufacturing processes cover 3D printing processes in which material is joined to a structure under computer control to create a three-dimensional object, where the structure is to be solidified, e.g. by sintering, to provide the macroscopic structure. Moreover, such metal additive manufacturing processes cover 3D printing processes, which do not require subsequent sintering, such as powder bed fusion or direct energy deposition processes. Examples of such powder bed fusion or direct energy deposition processes are laser beam, electron beam or plasma 3D printing processes.

The reactor system does not need a furnace and this reduces the overall reactor size considerably.

The electrically conductive material comprises Fe, Ni, Cu, Co, Cr, Al, Si or an alloy thereof. Such an alloy may comprise further elements, such as Mn, Y, Zr, C, Co, Mo or combinations thereof. Preferably, the electrically conductive material comprises Fe, Cr, Al or an alloy thereof. Such an alloy may comprise further elements, such as Si, Mn, Y, Zr, C, Co, Mo or combinations thereof. Preferably, the catalytically active material is particles having a size from 2 nm to 250 nm. Preferably, the conductors and the electrically conductive material are made of different materials than the electrically conductive material. The conductors may for example be of iron, nickel, aluminum, copper, silver or an alloy thereof. The ceramic coating is an electrically insulating material and will typically have a thickness in the range of around 100 μm , say 10-500 μm .

The electrically conductive material is advantageously a coherent or consistently intra-connected material in order to achieve electrical conductivity throughout the electrically conductive material, and thereby achieve thermal conductivity throughout the structured catalyst and in particular providing heating of the catalyst material. By the coherent or consistently intra-connected material it is possible to ensure uniform distribution of current within the electrically conductive material and thus uniform distribution of heat within the structured catalyst. Throughout this text, the term "coherent" is meant to be synonymous to cohesive and thus refer to a material that is consistently intra-connected or consistently coupled. The effect of the structured catalyst being a coherent or consistently intra-connected material is

6

that a control over the connectivity within the material of the structured catalyst and thus the conductivity of the electrically conductive material is obtained. It is to be noted that even if further modifications of the electrically conductive material are carried out, such as provision of slits within parts of the electrically conductive material or the implementation of insulating material within the electrically conductive material, the electrically conductive material is still denoted a coherent or consistently intra-connected material.

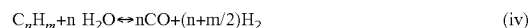
The gas flow over the structured catalyst may be axial or co-axial with the current path through the structured catalyst, perpendicular to the current path or have any other appropriate direction in relation to the current path.

The term "steam reforming" is meant to denote a reforming reaction according to one or more of the following reactions:



Reactions (i) and (ii) are steam methane reforming reactions, whilst reaction (iii) is the dry methane reforming reaction.

For higher hydrocarbons, viz. C_nH_m , where $n \geq 2$, $m \geq 4$, equation (i) is generalized as:



where $n \geq 2$, $m \geq 4$.

Typically, steam reforming is accompanied by the water gas shift reaction (v):



The term "steam methane reforming" is meant to cover the reactions (i) and (ii), the term "steam reforming" is meant to cover the reactions (i), (ii) and (iv), whilst the term "methanation" covers the reverse reaction of reaction (i). In most cases, all of these reactions (i)-(v) are at, or close to, equilibrium at the outlet from the reactor system.

The term "prereforming" is often used to cover the catalytic conversion of higher hydrocarbons according to reaction (iv). Prereforming is typically accompanied by steam reforming and/or methanation (depending upon the gas composition and operating conditions) and the water gas shift reaction. Prereforming is often carried out in adiabatic reactors but may also take place in heated reactors.

The steam reforming reaction is highly endothermic. High temperatures typically in excess of $800-850^\circ \text{C}$. are needed to reach acceptable conversions of the methane in the feed. A SMR consists of a number of tubes filled with catalyst pellets placed inside a furnace. The tubes are typically 10-13 meters long and will typically have an inner diameter between 80 and 160 mm. Burners placed in the furnace provide the required heat for the reactions by combustion of a fuel gas. A maximum average heat flux of 80000-90000 kcal/h/m² of inner tube surface is not uncommon. There is a general limitation to the obtainable heat flux due to mechanical constraints and the capacity is therefore increased by increasing the number of tubes and the furnace size. More details on the SMR type reactor system can be found in the art, e.g. "Synthesis gas production for FT synthesis"; Chapter 4, p. 258-352, 2004.

The term "electrically conductive" is meant to denote materials with an electrical resistivity in the range from: 10^{-5} to $10^{-8} \Omega\cdot\text{m}$ at 20°C . Thus, materials that are electrically conductive are e.g. metals like copper, silver, aluminum, chromium, iron, nickel, or alloys of metals. Moreover,

the term “electrically insulating” is meant to denote materials with an electrical resistivity above $10 \Omega \cdot \text{m}$ at 20°C ., e.g. in the range from 10^9 to $10^{25} \Omega \cdot \text{m}$ at 20°C .

When the reactor system comprises a heat insulation layer between the structured catalyst and the pressure shell, appropriate heat and electrical insulation between the structured catalyst and the pressure shell is obtained. The presence of heat insulating layer between the pressure shell and the structured catalyst assists in avoiding excessive heating of the pressure shell, and assists in reducing thermal losses to the surroundings. The temperatures of the structured catalyst may reach up to about 1300°C ., at least at some parts thereof, but by using the heat insulation layer between the structured catalyst and the pressure shell the temperature of the pressure shell can be kept at significantly lower temperatures of say 500°C . or even 100°C ., which is advantageous as typical construction steel materials typically are unsuitable for pressure bearing application at temperatures above 1000°C . Moreover, a heat insulating layer between the pressure shell and the structured catalyst assists in control of the electrical current within the reactor system, since heat insulation layer is also electrically insulating. The heat insulation layer could be one or more layers of solid material, such as ceramics, inert material, bricks or a gas barrier or a combination thereof. Thus, it is also conceivable that a purge gas or a confined gas constitutes or forms part of the heat insulation layer.

Moreover, it should be noted that the term “heat insulating material” is meant to denote materials having a thermal conductivity of about $10 \text{ W} \cdot \text{m}^{-1} \cdot \text{K}^{-1}$ or below. Examples of heat insulating materials are ceramics, bricks, fiber materials, alumina based materials, zirconia based materials and similar.

Advantageously, any relevant gaps between structured catalyst, the heat insulation layer, the pressure shell, and/or any other components inside the reactor system is filled with inert material, e.g. in the form of inert pellets. Such gaps are e.g. a gap between the lower side of the structured catalyst and the bottom of the pressure shell and a gap between the sides of the structured catalyst and the insulation layer covering the inner sides of the pressure shell. The inert material may e.g. be a ceramic material in the form of pellets or tiles. The inert material assists in controlling the gas distribution through the reactor system and in controlling the flow of the gas through the structured catalyst. Moreover, the inert material typically has a heat insulating effect.

The pressure shell suitably has a design pressure of between 2 bar and 30 bar. The actual operating pressure will be determined by the endothermic reaction, the size of the plants, among other aspects. As the hottest part of the reactor system is the electrically conductive material, which will be surrounded by heat insulation layer and within the pressure shell of the reactor system, the temperature of the pressure shell can be kept significantly lower than the maximum process temperature. This allows for having a relative low design temperature of the pressure shell of e.g. 700°C . or 500°C . or preferably 300°C . or 100°C . of the pressure shell whilst having maximum process temperatures of 400°C ., or even 900, or even 1100°C ., or even up to 1300°C . on the structured catalyst. Material strength is higher at the lower of these temperatures (corresponding to the design temperature of the pressure shell as indicated above). This offers advantages when designing the chemical reactor. Suitably, the pressure shell has a design pressure of between 30 bar and 200 bar, preferably between 80 and 180 bar.

The resistivity of the electrically conductive material is suitably between $10^{-5} \Omega \cdot \text{m}$ and $10^{-7} \Omega \cdot \text{m}$. A material with a

resistivity within this range provides for an efficient heating of the structured catalyst when energized with a power source. Graphite has a resistivity of about $10^{-5} \Omega \cdot \text{m}$ at 20°C ., kanthal has a resistivity of about $10^{-6} \Omega \cdot \text{m}$ at 20°C ., whilst stainless steel has a resistivity of about $10^{-7} \Omega \cdot \text{m}$ at 20°C . The electrically conductive material may for example be made of FeCr Alloy having a resistivity of ca. $1.5 \cdot 10^{-6} \Omega \cdot \text{m}$ at 20°C .

Typically, the pressure shell comprises an inlet for letting in process gas and an outlet for letting out product gas, wherein the inlet is positioned close to a first end of the pressure shell and the outlet is positioned close to a second end of the pressure shell, and wherein the at least two conductors both are connected to the structured catalyst at a position on the structured catalyst closer to the inlet than to the outlet. Hereby, the at least two conductors can be placed in the substantially colder part of the reactor system as the inlet gas will have lower temperature than the product gas, the electrically conductive material will be colder in the colder in the most upstream part of the material due to the heat consumed by the progress of the chemical reaction, and the feed gas fed led through the inlet may cool the at least two conductors before being heated by the heated structured catalyst further along the path of the gas over the heated structured catalyst. It is an advantage that the temperature of all electrically conducting elements except the electrically conductive material is kept down in order to protect the connections between the conductors and the structured catalyst. When the temperature of the conductors and other electrically conducting elements, except the electrically conductive material, is relatively low, less limitations on materials suitable for the conductors and other electrically conducting elements, except the electrically conductive material, exists. When the temperature of the electrically conducting elements increase, the resistivity thereof increases; therefore, it is desirable to avoid unnecessary heating of all other parts than the electrically conductive materials within the reactor system. The term “electrically conducting elements, except the electrically conductive material” is meant to cover the relevant electrically conducting elements arranged to connect the power supply to the structured catalyst.

It should be noted, that the system of the invention may include any appropriate number of power supplies and any appropriate number of conductors connecting the power supply/supplies and the electrically conductive material(s) of the structured catalyst. As mentioned above, a power supply is placed outside said pressure shell and is arranged to supply electrical power via electrical conductors connecting to said structured catalyst, allowing an electrical current to run through said macroscopic structure, thereby heating at least part of the structured catalyst.

Suitably, the at least two conductors are led through a pressure shell in a fitting so that the at least two conductors are electrically insulated from the pressure shell. The fitting may be, partly, of a plastic and/or ceramic material. The term “fitting” is meant to denote a device that allows for mechanically connecting two pieces of hardware in a pressure bearing configuration. Thereby, the pressure within the pressure shell may be maintained even though the at least two conductors are lead through it. Non-limiting examples of the fittings may be an electrically insulating fitting, a dielectric fitting, a power compression seal, a compression fitting or a flange. The pressure shell typically comprises side walls, end walls, flanges and possibly further parts. The term “pressure shell” is meant to cover any of these components.

The pressure shell may further comprise one or more inlets close to or in combination with at least one of the fittings in order to allow a cooling gas to flow over, around, close to or inside at least one conductor within said pressure shell. Hereby, the conductors are cooled and thus the temperature that the fitting experiences is kept down. If the cooling gas is not used, the conductors may be heated by the feed gas to the reactor system, resistance heating of conductor due to the applied current, and/or heat conduction from the structured catalyst. The cooling gas could e.g. be hydrogen, nitrogen, steam, carbon dioxide or mixtures thereof. The temperature of the cooling gas at entry into the pressure shell may be e.g. about 100° C. or 200° C. or 250° C. In an embodiment, the conductor(s) is (are) hollow so that the cooling gas may flow through the conductor(s) and cool it (them) from within. By keeping the temperature of the fitting low, e.g. at around 100-200° C., it is easier to have a leak tight configuration. Typically, a part of the feed gas, such as one of the reactants, is fed to the pressure shell as the cooling gas. In another embodiment, part of the feed gas or a gas with the same composition as the feed gas is used as cooling gas.

The reactor system may further comprise an inner tube in heat exchange relationship with the structured catalyst, where the inner tube is adapted to withdraw a product gas from the structured catalyst so that the product gas flowing through the inner tube or tubes is in heat exchange relationship with the gas flowing over the structured catalyst, but electrically separated from the structured catalyst. This is a layout which here is denoted a bayonet reactor system. In this layout the product gas within the inner tube assists in heating the process gas flowing over the structured catalyst. The electrical insulation between the inner tube and the structured catalyst could be gas in the form of a gap or distance between the inner tube and the structured catalyst or inert material loaded around the inner tube and the structured catalyst. The gas may pass through the structured catalyst in an up-flow or a down-flow direction.

The connection between the structured catalyst and the at least two conductors may be a mechanical connection, a welded connection, a brazed connection or a combination thereof. The structured catalyst may comprise terminals physically and electrically connected to the structured catalyst in order to facilitate the electrical connection between the electrically conductive material and the at least two conductors. The term "mechanical connection" is meant to denote a connection where two components are held together mechanically, such as by a threaded connection or by clamping, so that a current may run between the components.

The electrically conductive materials placed in an array of electrically conductive materials may be electrically connected to each other. The connection between the two or more electrically conductive materials may be by mechanical connection, clamping, soldering, welding or any combination of these connection methods. Each electrically conductive material may comprise terminals in order to facilitate the electrical connections. The two or more electrically conductive materials may be connected to the power supply in serial or parallel connection. The electrical connection between the two or more electrically conductive materials is advantageously coherent and uniform along the connection surface between the two or more electrically conductive materials, so that the two or more electrically conductive materials act as a single coherent or consistently intra-connected material; hereby, uniform electrical conductivity throughout the two or more electrically conductive

materials is facilitated. Alternatively, or additionally, the structured catalyst may comprise an array of electrically conductive materials that are not electrically connected to each other. Instead, two or more electrically conductive materials are placed together within the pressure shell, but not connected electrically to each other. In this case, the structured catalyst thus comprises electrically conductive materials connected in parallel to the power supply.

A ceramic coating, with or without catalytically active material, may be added directly to a metal surface of the electrically conductive material by wash coating. The wash coating of a metal surface is a well-known process; a description is given in e.g. Cybulski, A., and Moulijn, J. A., "Structured catalysts and reactors", Marcel Dekker, Inc, New York, 1998, Chapter 3, and references herein. The ceramic coat may be added to the surface of the electrically conductive material and subsequently the catalytically active material may be added; alternatively, the ceramic coat comprising the catalytically active material is added to the macroscopic structure or electrically conductive material. The ceramic coating may for example be an oxide comprising Al, Zr, Mg, Ce and/or Ca. Exemplary coatings are calcium aluminate or a magnesium aluminum spinel. Such a ceramic coating may comprise further elements, such as La, Y, Ti, K or combinations thereof. The ceramic coating is an electrically insulating material and will typically have a thickness in the range of around 100 µm, say 10-500 µm.

Extruding and sintering or 3D printing a macroscopic structure results in a uniformly and coherently shaped macroscopic structure, which can afterwards be coated with the ceramic coating.

The electrically conductive material and the ceramic coating may have been sintered in an oxidizing atmosphere in order to form chemical bonds between the ceramic coating and the electrically conductive material; this provides for an especially high heat conductivity between the electrically conductive material and the catalytically active material supported by the ceramic coating. Thereby, the structured catalyst is compact in terms of heat transfer to the active catalytic site, and a reactor system housing the structured catalyst may be compact and limited mainly by the rate of the chemical reaction. There is no heat transfer from outside the pressure shell to the structured catalyst as would be the case through the tube walls to the catalyst within the tubes for the SMRs used in the art of steam reforming.

In an embodiment, the structured catalyst has at least one electrically insulating part arranged to increase the current path between the conductors to a length larger than the largest dimension of the structured catalyst. The provision of a current path between the conductors larger than the largest dimension of the structured catalyst may be by provision of electrically insulating part(s) positioned between the conductors and preventing the current running through some part of the structured catalyst. Such electrically insulating parts are arranged to increase the current path and thus increase the resistance through the structured catalyst. Hereby, the current path through the structured catalyst can be e.g. more than 50%, 100%, 200%, 1000%, or even 10000% longer than the largest dimension of the structured catalyst.

Moreover, such electrically insulating parts are arranged to direct the current from one conductor, which is closer to the first end of the structured catalyst than to the second end, towards the second end of the structured catalyst and back to a second conductor closer to the first end of the structured catalyst than to the second end. Preferably, the current is

arranged to run from the first end of the structured catalyst to the second and back to the first end. As seen in the figures, the first end of the structured catalyst is the top end thereof. The arrow indicated "z" in FIGS. 5-7 indicates a z-axis along the length of the structured catalyst. The principal current path throughout the structured catalyst will have a positive or negative value of z-coordinate of the accompanied current density vector along most of the length of the current path. By principal current path is meant the path of the electrons through a macroscopic structure of the structured catalyst with the highest current density. The principal current path can also be understood as the path having the minimum length through the macroscopic structure of the structured catalyst. Seen geometrically, the principal current path can be quantified as the largest current density vector within a plane perpendicular to the gas flow direction of a coherent section of the macroscopic structure. At the bottom of the structured catalyst, as shown in the figures, the current will turn, and here the z-coordinate of the accompanied current density vector will be zero.

As used herein, the term coherent section is meant to denote a cross-section area of the macroscopic structure wherein all walls of the coherent section are geometrically connected to one or more other walls of the coherent section within the same plane.

In an embodiment, the structured catalyst has at least one electrically insulating part arranged to direct a current through the structured catalyst in order to ensure that for at least 70% of the length of said structured catalyst, a current density vector of a principal current path has a non-zero component value parallel to the length of said structured catalyst. Thus, for at least 70% of the length of the structured catalyst, the current density vector will have a positive or negative component value parallel to the length of the structured catalyst. Thus, for at least 70%, e.g. for 90% or 95%, of the length of structured catalyst, viz. along the z-axis of the structured catalyst as seen in FIGS. 5 to 10, the current density vector of a principal current path will have a positive or negative value along the z-axis. This means that the current is forced from the first end of the structured catalyst towards the second end, and subsequently is forced towards the first end again. The temperature of the gas entering the first end of the structured catalyst and the endothermic steam reforming reaction taking place over the structured catalyst absorbs heat from the structured catalyst. For this reason, the first end of the structured catalyst remains colder than the second end, and by ensuring that the current density vector of the principal current path has a non-zero component value parallel to the length of said structured catalyst, this takes place with a substantially continuously increasing temperature profile, which gives a controllable reaction front. In an embodiment the current density vector has a non-zero component value parallel to the length of said structured catalyst in 70% of the length of said structured catalyst, preferably 80%, more preferably 90%, and even more preferably 95%. It should be noted that the term "the length of the structured catalyst" is meant to denote the dimension of the structured catalyst in the direction of the gas flow. In the structured catalysts as shown in the figures, the length is the longitudinal direction, viz. the longest dimension thereof. This is indicated by the arrow denote z in some of the figures.

Non-limiting examples of insulating parts are cuts, slits, or holes in the structure. Optionally, a solid insulating material such as ceramics in cuts or slits in the structure can be used. In a case where the solid insulating material is a porous ceramic material, the catalytically active material

may advantageously be incorporated in the pores, by e.g. impregnation. A solid insulating material within a cut or slit assists in keeping the parts of the structured catalyst on the sides of the cut or slit from each other. As used herein, the term "largest dimension of the structured catalyst" is meant to denote the largest inner dimension of the geometrical form taken up by the structured catalyst. If the structured catalyst is box-formed, the largest dimension would be the diagonal from one corner to the farthest corner, also denoted the space diagonal.

It should be noted that even though the current through the structured catalyst may be arranged to twist or wind its way through the structured catalyst due to the electrically insulating parts arranged to increase the current path, the gas passing through the reactor system is inlet at one end of the reactor system, passes over the structured catalyst once before being outlet from the reactor system. Inert material is advantageously present in relevant gaps between the structured catalyst and the rest of the reactor system to ensure that the gas within the reactor system passes over the structured catalyst and the catalyst material herein.

The length of the gas passage through the structured catalyst is suitably less than the length of the passage of current from one electrode through the structured catalyst and to the next electrode. The ratio of the length of the gas passage to the length of the current passage may be less than 0.6, or 0.3, 0.1, or even down to 0.002.

Typically, the structured catalyst has electrically insulating parts arranged to make the current path through the structured catalyst a zigzag path. Here, the terms "zigzag path" and "zigzag route" is meant to denote a path that has corners at variable angles tracing a path from one conductor to another. A zigzag path is for example a path going upwards, turning, and subsequently going downwards. A zigzag path may have many turns, going upwards and subsequently downwards many times through the structured catalyst, even though one turn is enough to make the path a zigzag path.

It should be noted that the insulating parts arranged to increase the current path are not necessarily related to the ceramic coating on the electrically conductive material; even though this ceramic coating is also considered electrically insulating, it does not change the length of the current path between the conductors connected to the electrically conductive material.

The macroscopic structure may have a plurality of parallel channels, a plurality of non-parallel channels and/or a plurality of labyrinthic channels, where the channels have walls defining the channels. Thereby, several different forms of the macroscopic structure can be used as long as the surface area of the structured catalyst exposed to the gas is as large as possible. In a preferred embodiment, the macroscopic structure has parallel channels, since such parallel channels render a structured catalyst with a very small pressure drop. In a preferred embodiment, parallel longitudinal channels are skewed in the longitudinal direction of the macroscopic structure. In this way, molecules of the gas flowing through the macroscopic structure will mostly tend to hit a wall inside the channels instead of just flowing straight through a channel without being in contact with a wall. The dimension of the channels should be appropriate in order to provide a macroscopic structure with a sufficient resistivity. For example, the channels could be quadratic (as seen in cross section perpendicular to the channels) and have a side length of the squares of between 1 and 3 mm; however, channels having a maximum extent in the cross section of up to about 4 cm are conceivable. The walls may e.g. have a

13

thickness of between 0.2 and 2 mm, such as about 0.5 mm, and the ceramic coating supported by the walls has a thickness of between 10 μm and 500 μm , such as between 50 μm and 200 μm , such as 100 μm . In another embodiment, the macroscopic structure of the structured catalyst is cross-

corrugated. In general, when the macroscopic structure is extruded or 3D printed, the pressure drop from the inlet to the outlet of the reactor system may be reduced considerably compared to a reactor where the catalyst material is in the form of pellets.

Suitably, the reactor system further comprises a bed of a second catalyst material upstream the structured catalyst within the pressure shell. Here, the term "upstream" is seen from the flow direction of the feed gas. Thus, the term "upstream" is here meant to denote that the feed gas is directed through the bed of second catalyst material prior to reaching the structured catalyst. This provides for a situation where the second catalyst material can be arranged for pre conditioning the feed stream. No specific heating needs to be provided to the bed of second catalyst material; however, the bed of second catalyst material may be heated indirectly if it is in close proximity to the structured catalyst. Alternatively, the second catalyst material may be heated. In order to clarify the terminology used here, it is noted that the term "structured catalyst" may also be denoted "a first catalyst material" to distinguish it from the second and/or third and/or fourth catalyst material.

The reactor system may further comprise a third catalyst material in the form of catalyst pellets, extrudates or granulates loaded into the channels of the macroscopic structure. In this embodiment, the reactor system will thus have a catalytically active material in the coating of the macroscopic structure as well as a third catalyst material in the form catalyst pellets, extrudates or granulates within the channels of the macroscopic structure. The pellets are e.g. prepared in a dimension to loosely match the size of channels to form a single string of pellets stacked upon each other within a channel of the macroscopic structure. Alternatively, the pellets, extrudates or granulates may be prepared in a dimension significantly smaller than the channel size to form a packed bed inside each channel. As used herein, the term "pellet" is meant to denote any well-defined structure having a maximum outer dimension in the range of millimeters or centimeters, while "extrudate" and "granulate" are meant to define a catalyst material with a maximum outer dimension defined within a range.

A bed of fourth catalyst material may be placed within the pressure shell and downstream the structured catalyst. Such fourth catalyst material may be in the form of catalyst pellets, extrudates or granulates.

Therefore the first, second, third, and fourth catalyst material may be catalyst materials suitable for the steam reforming reaction. In a configuration where a combination of the second, third, and fourth catalyst material is included in the reactor system, the catalyst of each catalyst material can be different.

The geometric surface area of the macroscopic structure may be between 100 and 3000 m^2/m^3 , such as between 500 and 1100 m^2/m^3 . Typically, the material of the macroscopic structure is chosen as a material arranged to supply a heat flux of 500 W/m^2 to 50000 W/m^2 by resistance heating of the material. Preferably, resistance heating of the material supplies a heat flux of between 5 kW/m^2 and 12 kW/m^2 , for example between 8 kW/m^2 and 10 kW/m^2 . The heat flux is given as heat per geometric surface area of the surface exposed to the gas.

14

The predetermined temperature range of the gas exiting the pressure shell/the reactor system is the range from 200 to 1300° C., depending on the endothermic reaction facilitated. The product gas outlet temperature from the structured catalyst is measured directly beneath or on the most downstream surface of the structured catalyst. Measuring technology can be thermocouples (by voltage drop), resistance temperature detectors or infrared detection. The measuring point can be separated from the structured catalyst and be embedded in downstream inert/catalyst, or be directly on the surface with an insulating surface coverage.

In order to control the temperature of a reaction, the heat added/removed from a reactor system needs to be balanced against the heat consumed/produced by the chemical reaction. The addition/removal of heat needs to be balanced against the rate of reaction and especially the approach to equilibrium as defined by β , where β is the ratio between the reaction quotient and the equilibrium constant of a reaction. A value of β approaching 1 means the reaction mixture is close to equilibrium and values approaching 0 means the reaction mixture is far from equilibrium. In general, it is desirable to have as high a rate of reaction as possible, which is achieved at a low β , as long as the temperature can be sufficiently controlled in parallel by balancing the energy added.

In the case of the endothermic steam methane reforming reaction, heat needs to be added to ensure the reaction continues to proceed as otherwise the reaction will be equilibrated and the β value will approach 1 and the reaction will slow down. However, on the other side, it is undesirable if the temperature increases faster than the rate of reaction can follow as exposing unconverted hydrocarbons to high temperatures can result in carbon formation. A good way to follow this behavior is by the approach to equilibrium. The approach to equilibrium of the steam reforming reaction is found by initially calculating the reaction quotient (Q) of the given gas as:

$$Q = \frac{y_{\text{CO}} \cdot y_{\text{H}_2}^3}{y_{\text{CH}_4} \cdot y_{\text{H}_2\text{O}}} \cdot P^2$$

Here y_j is the molar fraction of compound j , and P is the total pressure in bar. This is used to determine the equilibrium temperature (T_{eq}) at which the given reaction quotient is equal to the equilibrium constant:

$$Q = K_{SMR}(T_{eq})$$

where K_{SMR} is the thermodynamic equilibrium constant of the steam methane reforming reaction. The approach to equilibrium of the steam methane reforming ($\Delta T_{app,SMR}$) reaction is then defined as:

$$T_{app,SMR} = T - T_{eq}$$

Where T is the bulk temperature of the gas surrounding the catalyst material used, such as the structured catalyst.

To ensure good performance of a steam reforming catalyst, it is desirable that the catalyst continuously works towards decreasing $\Delta T_{app,SMR}$. Classically, large scale industrial SMRs have been designed to obtain an approach to equilibrium of 5-20° C. at the outlet thereof.

With the current invention, it is possible to control the heat flux and match this directly to the kinetic performance of the structured catalyst, as these are independent to some extent.

The structured catalyst within said reactor system suitably has a ratio between the area equivalent diameter of a

horizontal cross section through the structured catalyst and the height of the structured catalyst in the range from 0.1 to 2.0. The area equivalent diameter of the cross section through the reactor system is defined as the diameter of a circle of equivalent area as the area of the cross section. When the ratio between the area equivalent diameter and the height of the structured catalyst is between 0.1 and 2.0, the pressure shell housing the structured catalyst may be relatively small compared to other reactor systems for endothermic reactions such as the current SMRs.

Typically, the gas flows through the reactor system in an upflow or downflow direction, so that the gas flows through channels in the structured catalyst along the height thereof. When the structured catalyst comprises a number of or an array of macroscopic structures, the individual macroscopic structures within the array may be placed side by side, on top of each other or in a combination thereof. It is stressed that, when the structured catalyst comprises more than one macroscopic structures, the dimensions of the structured catalyst are the dimensions of the more than one macroscopic structures. Thus, as an example, if the structured catalyst comprises two macroscopic structures, each having the height h , put on top of each other, the height of the structured catalyst is $2h$.

The volume of the structured catalyst is chosen in consideration of the desired feed conversion and/or temperature out of the reactor system correlated to the heat generation capacity of the electrically conductive material.

Suitably, the height of the reactor system is between 0.5 and 7 m, more preferably between 0.5 and 3 m. Exemplary values of the height of the reactor system is a height of less than 5 meters, preferably less than 2 m or even 1 m. The dimensions of the reactor system and of the structured catalyst within the reactor system are correlated; of course, the pressure shell and heat insulation layer render the reactor system somewhat larger than the structured catalyst itself. For comparison, industrial scale SMRs are typically constructed of catalyst tubes having a length of 10 m or above to maximize external surface area of the tubes. The present invention is advantageous in that such confinements in the design of the reactor system are superfluous.

The reaction system described above is—in contrast to fired reformers—not a segregated system. As heating is not transferred across a pressure bearing wall, the risk of mechanical failure is not high. From a catalyst point of view, temporary overheating is not a problem, it will just produce a very hot synthesis gas. This means that start-up is fast in comparison and in practice the eSMR can be started by applying a given voltage and then the system will work towards a thermal equilibration to reach steady state without any additional operator input.

In a main aspect, therefore, a method for rapidly switching a metal-catalysed steam methane reforming reaction of a feed gas comprising hydrocarbons in a reactor system from a first steady-state reaction condition (A) to a second steady-state reaction condition (B) or vice-versa; is therefore provided.

Reaching a steady state condition is defined as when central process parameters (such as feed flow, outlet temperature, and reactant conversion) have reached a value within $\pm 15\%$ of the average process value for the given process parameter for the subsequent hour.

A condition of the invention, A or B, involves a state where the catalyst of the system is heated by an electrical power balanced to heat the product gas outlet temperature from the structured catalyst to a temperature between 300 and 1300° C. at a pressure between 5 barg and 150 barg with

a feedstock comprising hydrocarbons (such as methane, ethane, or propane), hydrogen, and steam, and any of carbon monoxide, carbon dioxide, nitrogen, argon, or oxygen in a total flow rate of 300 Nm³/h to 100 000 Nm³/h. When the feedstock passes the monolith, it will react towards equilibration of the steam reforming and water gas shift reaction.

In an embodiment of the invention, the method includes an initial reaction condition A where the feedstock consists of 8.7% CO₂, 0.2% N₂, 21.5% CH₄, 3.7% H₂, 0.1% CO, and 65.7% H₂O in a total flow of 9757 Nm³/h having a temperature of 415° C. at a pressure of 27.0 barg. Supplying a first electrical power of 725 kW generates an almost equilibrated gas composed of 9.4% CO₂, 0.2% N₂, 20.0% CH₄, 7.9% H₂, 0.2% CO, and 62.2% H₂O in a total flow of 9978 Nm³/h having a temperature of 500° C. at a pressure of 26.5 barg. Switching to condition B over a period of about 90 min while applying a second electrical power of 9710 kW generates an almost equilibrated gas composed of 6.1% CO₂, 0.2% N₂, 0.5% CH₄, 46.8% H₂, 14.8% CO, and 31.7% H₂O in a total flow of 13830 Nm³/h having a temperature of 990° C. at a pressure of 26.5 barg.

In an embodiment of the invention, the method includes an initial reaction condition A where the feedstock consists of 12.6% CO₂, 0.3% N₂, 33.9% CH₄, 2.5% H₂, 0.1% CO, and 50.6% H₂O in a total flow of 17488 Nm³/h having a temperature of 405° C. at a pressure of 77.0 barg. Supplying a first electrical power of 19.3 MW generates an almost equilibrated gas composed of 5.1% CO₂, 0.2% N₂, 6.5% CH₄, 47.5% H₂, 19.8% CO, and 21.0% H₂O in a total flow of 25998 Nm³/h having a temperature of 1005° C. at a pressure of 75.5 barg. Switching to condition B over a period of about 130 min while increasing the feedstock total flow to 43323 Nm³/h and applying a second electrical power of 46.9 MW generates an almost equilibrated gas composed of 5.3% CO₂, 0.2% N₂, 7.0% CH₄, 46.7% H₂, 19.3% CO, and 21.5% H₂O in a total flow of 64408 Nm³/h having a temperature of 995° C. at a pressure of 75.5 barg.

The term “vice versa” is used to mean that the method applies equally when switching from the first reaction condition (A) to the second reaction condition (B) as when switching from the second reaction condition (B) to the first reaction condition (A). Notably, a switch from condition A to B is considered completed when the process values of the system have reached within 85% of steady state conditions.

The reactor system is as described above; i.e. it comprises a pressure shell housing a structured catalyst arranged to catalyze steam reforming of a feed gas comprising hydrocarbons, said structured catalyst comprising a macroscopic structure of an electrically conductive material, said macroscopic structure supporting a ceramic coating, where said ceramic coating supports a catalytically active material and wherein said reactor system is provided with heat insulation between said structured catalyst and said pressure shell. All details described above in relation to the reactor system are relevant for the present technology.

The method of the invention comprises the steps of:

in said first steady-state reaction condition (A):

supplying said feed gas to the reactor system in a first total flow, and

supplying a first electrical power via electrical conductors connecting an electrical power supply placed outside said pressure shell to said structured catalyst, thereby allowing a first electrical current to run through said electrically conductive material,

thereby heating at least part of the structured catalyst to a first temperature at which said feed gas is converted to a first product gas mixture over said structured catalyst

17

under said first steady-state reaction conditions (A); and said first product gas is outlet from the reactor system;
 and, in said second steady-state reaction condition (B):
 supplying said feed gas to the reactor system in a second total flow,
 supplying a second electrical power via electrical conductors connecting an electrical power supply placed outside said pressure shell to said structured catalyst, thereby allowing a second electrical current to run through said electrically conductive material,
 thereby heating at least part of the structured catalyst to a second temperature; at which said feed gas is converted to a second product gas mixture over said structured catalyst under said second steady-state reaction conditions (B); and said second product gas is outlet from the reactor system;

To achieve the first and second steady-state reaction conditions (A) and (B), the second electrical power is higher than said first electrical power; and/or said second total flow is higher than said first total flow.

Notably, an increase in total flow will increase the input of cool feed gas, thus cooling the structured catalyst, and reducing the reactivity so that second steady-state reaction condition (B) is achieved. A significant change in flow will change the energy required for the process.

A change in total flow may include a change in total flow with no compositional change or a change in the composition, such as increasing the steam flow or changing part of the feedstock.

In one embodiment, the ratio of total gas feed flow in said first reaction condition A to said second reaction condition B (A:B) is at least 1:10. Switching between condition A and B consequently allows for significant increased/decreased production of product gas. This is advantageous when the invention is used for e.g. energy storage where excess electric energy from the energy grid is available and in this way can be stored as chemical energy, or vice versa for increasing availability of electric energy in the grid when it is needed elsewhere. Additionally, the embodiment allows for using the invention to supply large amounts of product gas in periods where downstream processes demands it, while having the invention operating in a standby condition otherwise. This is advantageously if there is no continuous demand for the product gas.

In another embodiment, the product gas outlet temperature from the structured catalyst in reaction condition B is between 50° C. to 800° C. higher, such as between 100° C. to 500° C. higher, preferably between 150° C. to 400° C. higher, than the product gas outlet temperature from the structured catalyst in reaction condition A. This allows for rapidly starting up the reactor system from a cold state to operating conditions. This is advantageously in the situation of system start-up, where the start-up procedure involves steps including:

Heating process equipment in a non-condensing gas to a temperature above the condensation point of the steady state conditions of the plant at full operating capacity,
 Pressurising the feed gas constituents,
 Feeding feed gas constituents to the reactor system while applying a first electrical power,
 Switching to a higher operating temperature by applying a second electrical power.

In this way, all steps of the start-up procedure are relatively fast. In the closest prior art, of e.g. a tubular reformer,

18

the step of switching to higher operating temperature is a very slow process and is time-wise an order of magnitude greater than the other steps.

The product gas outlet temperature from the structured catalyst in reaction condition B is typically no more than 50° C. higher than the product gas outlet temperature from the structured catalyst in reaction condition A. This allows for rapidly changing the between condition A and B, without significantly changing the product gas composition from the system. In this way, the demand for the product gas for downstream processes of the reactor system can easily be supplied in different quantities without interfering significantly in the chemical environment of these.

In one embodiment, the switch between reaction condition A and B includes a gradual change of the total gas feed flow from said first total flow to said second total flow and simultaneous gradual change of the applied electrical potential over said electrically conductive material from said first to said second electrical power. In this way, the product gas composition can be held almost constant also during the transition stage. In an embodiment, the gradual changes are made in such a way where the flow is increased in small steps while increasing the electrical power to maintain an almost constant product gas outlet temperature from the structured catalyst.

In an embodiment, the reactor system further comprises a control system arranged to control the electrical power supply to ensure that the temperature of the gas exiting the pressure shell lies in a predetermined range and/or to ensure that the conversion of the feed gas lies in a predetermined range. The control of the electrical power supply is the control of the electrical output from the power supply. The control of the electrical power supply may e.g. be carried out as a control of the voltage and/or current from the electrical power supply, as a control of whether the electrical power supply is turned on or off or as a combination hereof. The power supplied to the structured catalyst can be in the form of alternating current or direct current.

According to one embodiment, a proportional-integral-derivative (PID) controller controls the electrical potential based on feedback reading of the process value of product gas outlet temperature from the structured catalyst.

The method described herein allows rapid switching between conditions A and B. Suitably, therefore, the switch between reaction conditions A and B takes place over a period of less than 3 hours, such as less than 2 hours, such as less than 60 min, preferably less than 30 min, and even more preferably less than 15 min. This is far quicker than the corresponding switch between reaction conditions A and B which would be possible with a fired reformer.

In one embodiment, the switch between reaction condition A and B involves supplying a second electrical power to the structured catalyst. This suitably occurs while keeping the total flow essentially constant.

In one aspect, the switch between reaction condition A and B comprises a transition state between said reaction conditions A and B; said transition state comprising a first period in which the electrical power is switched off, followed by a second period in which said second electrical power of condition B is supplied to the structured catalyst. This allows for faster establishment of a steady state.

In one aspect, the switch between reaction condition A and B comprises a transition state between said reaction conditions A and B; said transition state comprising a first period in which a third electrical power is supplied to the structured catalyst, followed by a second period in which said second electrical power of condition B is supplied to the

19

structured catalyst, said third electrical power being higher than the second electrical power. This allows for faster establishment of a steady state.

The process may comprise further steps carried out on the product gas, such as purification, pressurization, heating, cooling, etc. to provide the final product gas for an application downstream the reactor system of this invention.

It should be noted that the feed gas may comprises individual feed gasses and that the step of pressurizing the feed gas may comprise pressurizing individual feed gasses individually. Moreover, it should be noted that the order in which the steps of the process are written are not necessarily the order in which the process steps take place, in that two or more steps may take place simultaneously, or the order may be different than indicated above.

In an embodiment, the process comprises the step of pressurizing the gas upstream the pressure shell to a pressure of up to at least 2 bar. The chosen operating pressure is defined by the endothermic reaction and the integration of the reactor in the surrounding process steps.

In an embodiment of the process according to the invention, the temperature of the feed gas let into the reactor system is between 100° C. and 700° C. However, in all embodiments the temperature and the pressure of the feed gas are adjusted to ensure that the feed gas is above the dew point.

In an embodiment of the process of the invention, the structured catalyst is heated so that the maximum temperature of the structured catalyst lies between 200° C. and 1300° C. The used temperature will be dependent on the endothermic reaction. The maximum temperature of the structured catalyst depends upon the material of the electrically conductive material; thus, if the electrically conductive material is of FeCr Alloy, which melts at a temperature of between 1380° C. and 1490° C. (depending on the actual alloy), the maximum temperature should be somewhat below the melting point, such as at about 1300° C. if the melting point of the electrically conductive material is at about 1400° C., as the material will become soft and ductile when approaching the melting point. The maximum temperature may additionally be limited by the durability of the catalyst material, the coating and the catalytically active material.

In an embodiment the process according to the invention further comprises the step of inletting a cooling gas through an inlet through the pressure shell in order to allow a cooling gas to flow over at least one conductor and/or fitting. The cooling gas may advantageously be hydrogen, nitrogen, steam, carbon dioxide or any other gas suitable for cooling the area or zone around the at least one conductor. A part of the feed gas may be fed to the pressure shell as the cooling gas.

In an embodiment according to the invention, the process further comprises the step of inletting a cooling gas through an inlet through the pressure shell in order to allow a cooling gas to flow over at least one conductor and/or fitting. The cooling gas may be any appropriate gas; examples of such gasses are hydrogen, nitrogen, steam, carbon dioxide or mixtures thereof. The cooling gas may flow through the conductor(s) and cool it (them) from within; in this case, the conductor(s) need(s) to be hollow to accommodate the cooling gas flowing within it/them.

Steam reforming of hydrocarbons takes place according to reaction (i)-(v), above. The catalyst material for the reaction may be Ni/Al₂O₃, Ni/ZrO₂, Ni/MgAl₂O₃, Ni/CaAl₂O₃, Ru/MgAl₂O₃, or Rh/MgAl₂O₃. The catalytically active material may be Ni, Ru, Rh, Ir, or a combination

20

thereof, while the ceramic coating may be Al₂O₃, ZrO₂, MgAl₂O₃, CaAl₂O₃, or a combination therefore and potentially mixed with oxides of Y, Ti, La, or Ce. The maximum temperature of the reactor may be between 850-1300° C. The pressure of the feed gas may be 15-180 bar, preferably about 25 bar. In an embodiment said macroscopic structure is made of an alloy of Fe Cr Al, supporting a ceramic coating of a ZrO₂ and MgAl₂O₄ mixture, with nickel as catalytically active material.

The following is a detailed description of embodiments of the invention depicted in the accompanying drawings. The embodiments are examples and are in such detail as to clearly communicate the invention. However, the amount of detail offered is not intended to limit the anticipated variations of embodiments; but on the contrary, the intention is to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the present invention as defined by the appended claims.

DETAILED DESCRIPTION OF THE FIGURES

Throughout the Figures, like reference numbers denote like elements.

FIG. 1a shows a cross section through an embodiment of a reactor system 100 according to the invention. The reactor system 100 comprises a structured catalyst 10, arranged as an array of macroscopic structures 5. Each macroscopic structure 5 in the array is coated with a ceramic coating impregnated with catalytically active material. The reactor system 100 moreover comprises conductors 40, 40' connected to a power supply (not shown in the Figures) and to the structured catalyst 10, viz. the array of macroscopic structures. The conductors 40, 40' are led through the wall of a pressure shell 20 housing the structured catalyst and through insulating material 30 on the inner side of the pressure shell, via fittings 50. The conductors 40' are connected to the array of macroscopic structures 5 by conductor contact rails 41.

In an embodiment, the electrical power supply supplies a voltage of 26V and a current of 1200 A. In another embodiment, the electrical power supply supplies a voltage of 5V and a current of 240 A. The current is led through electrical conductors 40, 40' to conductor contact rails 41, and the current runs through the structured catalyst 10 from one conductor contact rail 41, e.g. from the conductor contact rail seen to the left in FIG. 1a, to the other conductor contact rail 41, e.g. the conductor contact rail seen to the right in FIG. 1a. The current can be both alternating current, and e.g. run alternating in both directions, or direct current and run in any of the two directions.

The macroscopic structures 5 are made of electrically conductive material. Especially preferred is the alloy kant-hal, also known as FeCr Alloy, consisting of aluminum, iron and chrome. The ceramic coating, e.g. an oxide, coated onto the structure catalysts 5 is impregnated with catalytically active material. The conductors 40, 40' are made in materials like iron, aluminum, nickel, copper or alloys thereof.

During operating, a feed gas enters the reactor system 100 from above as indicated by the arrow 11 and exits the reactor system from the bottom thereof as indicated by the arrow 12.

FIG. 1b shows the reactor system 100 of FIG. 1a with a part of the pressure shell 20 and heat insulation 30 layer removed and FIG. 2 is an enlarged view of a part of the reactor system 100. In FIGS. 1b and 2 the connections between conductors 40' and conductor contact rails 41 are shown more clearly than in FIG. 1a. Moreover, it is seen that the conductors 40 are led through the walls of the pressure

shell in a fitting 50, and that the one conductor 40 is split up into three conductors 40' within the pressure shell. It should be noted, that the number of conductors 40' may be any appropriate number, such as smaller than three or even larger than three.

In the reactor system shown in FIGS. 1a, 1b and 2, the conductors 40, 40' are led through the wall of a pressure shell 20 housing the structured catalyst and through insulating material 30 on the inner side of the pressure shell, via fittings 50. Feed gas for steam reforming is inlet into the reactor system 100 via an inlet in the upper side of the reactor system 100 as shown by the arrow 11, and reformed gas exits the reactor system 100 via an outlet in the bottom of the reactor system 100 as shown by the arrow 12. Moreover, one or more additional inlets (not shown in FIGS. 1a to 2) advantageously exist close to or in combination with the fittings 50. Such additional inlets allow a cooling gas to flow over, around, close to, or inside at least one conductor within the pressure shell to reduce the heating of the fitting. The cooling gas could e.g. be hydrogen, nitrogen, steam, carbon dioxide or mixtures thereof. The temperature of the cooling gas at entry into the pressure shell may be e.g. about 100° C.

In the reactor system 100 shown in FIGS. 1a to 2, inert material (not shown in FIGS. 1a-2) is advantageously present between the lower side of the structured catalyst 10 and the bottom of the pressure shell. Moreover, inert material is advantageously present between the outer sides of the structured catalyst 10 of macroscopic structures 5 and the insulating material 30. Thus, one side of the insulating material 30 faces the inner side of the pressure shell 20 and the other side of the insulating material 30 faces the inert material. The inert material is e.g. ceramic material and may be in the form of pellets. The inert material assists in controlling the pressure drop across the reactor system 100 and in controlling the flow of the gas through the reactor system 100, so that the gas flows over the surfaces of the structured catalyst 10.

FIGS. 3a and 3b show schematic cross sections through an embodiment of the inventive reactor system 100', 100" comprising a structured catalyst 10'. The structured catalyst 10' may consist of a single macroscopic structure with ceramic coating supporting catalytically active material or it may contain two or more macroscopic structures. Each of the reactor systems 100', 100" comprises a pressure shell 20 and a heat insulation layer 80 between the structured catalyst 10' and the pressure shell 20. Inert material 90 can be used to fill the gap between the structured catalyst 10' and the heat insulation layer or the pressure shell 20. In FIGS. 3a and 3b, the inert material 90 is indicated by dotted area; the inert material 90 may be in any appropriate form, e.g. in the form of inert pellets, and it is e.g. of ceramic material. The inert material 90 assists in controlling the pressure drop through the reactor system and in controlling the flow of the gas through the reactor system. Moreover, the inert material typically has a heat insulating effect.

From FIGS. 3a and 3b it is seen that the reactor systems 100', 100" further comprise an inner tube 15 in heat exchange relationship with the structured catalyst 10'. The inner tube 15 is adapted to withdraw a product gas from the structured catalyst 10' so that the product gas flowing through the inner tube or tubes is in heat exchange relationship with the gas flowing over the structured catalyst; however, the inner tube 15 is electrically insulated from the structured catalyst 10' by either a heat insulation layer 80, inert material 90, a gap, or a combination. This is a layout which is denoted a bayonet reactor system. In this layout, the

product gas within the inner tube assists in heating the process gas flowing over the macroscopic structure. In the layouts shown in FIGS. 3a and 3b, the feed gas enters the reactor system 100', 100" as indicated by the arrow 11, and continues into the structured catalyst 10' as indicated by the arrows 13. During the passage of the feed gas over the structured catalyst 10', it undergoes the steam reforming reaction. The gas exiting the structured catalyst 10' is at least partly reformed. The at least partly reformed gas flows from the structured catalyst 10' into the inner tube 15 as indicated by the arrows 14, and exits the inner tube as indicated by the arrows 12. Even though the heat insulation layer 80 is present between the inner tube 15 and the structured catalyst 10', some heat transfer will take place from the gas within the inner tube 15 and the gas within the structured catalyst 10' or upstream the structured catalyst 10'. In the embodiments shown in FIGS. 3a and 3b, the feed gas flow downwards through the structured catalyst 10' and upwards through the inner tube 15; however, it is conceivable that the configuration was turned upside-down so that the feed gas would flow upwards through the structured catalyst 10' and downwards through the inner tube 15.

FIGS. 4 and 5 show an embodiment of a structured catalyst comprising an array of macroscopic structures as seen from above and from the side, respectively. FIG. 4 shows a structured catalyst 10 comprising an array of macroscopic structure 5 seen from above, viz. as seen from the arrow 11 in FIGS. 1a and 1b. The array has 6 rows, viz. 1a, 1b, 1c, 1d, 1e and 1f, of five macroscopic structures 5. The macroscopic structures 5 in each row are connected to its neighboring macroscopic structure (s) in the same row and the two outermost macroscopic structures in each row are connected to a conductor contact rail 41. The neighboring macroscopic structures 5 in a row of macroscopic structures are connected to each other by means of a connection piece 3.

FIG. 5 shows the structured catalyst 10 having an array of macroscopic structures 5 of FIG. 4 seen from the side. From FIG. 5, it can be seen that each macroscopic structure 5 extends longitudinally perpendicular to the cross section seen in FIG. 4. Each macroscopic structure 5 has a slit 60 cut into it along its longitudinal direction (see FIG. 5). Therefore, when energized by the power source, the current enters the array of macroscopic structures 5 via a conductor contact rail 41, is led through the first macroscopic structure 5 downwards until the lower limit of the slit 60 and is subsequently led upwards towards a connection piece 3. The current is led via a corresponding zigzag path, downwards and upwards, through each macroscopic structure 5 in each row 1a-1f of macroscopic structures 5 in the array 10. This configuration advantageously increases the resistance over the structured catalyst 10.

FIG. 6 shows a structured catalyst 10 according to the invention in a perspective view. The structured catalyst 10 comprises a macroscopic structure that is coated with a ceramic coating impregnated with catalytically active material. Within the structured catalyst are channels 70 extending along the longitudinal direction (shown by the arrow indicate 'h' in FIG. 6) of the macroscopic structure 5; the channels are defined by walls 75. In the embodiment shown in FIG. 6, the walls 75 define a number of parallel, square channels 70 when seen from the direction of flow as indicated by the arrow 12. The structured catalyst 10 has a substantially square perimeter when seen from above, defined by the edge lengths e1 and e2. However, the perimeter could also be circular or another shape.

23

The walls **75** of the structured catalyst **10** are of extruded or 3D printed material coated with a ceramic coating, e.g. an oxide, which has been coated onto the macroscopic structure. In the Figures, the ceramic coating is not shown. The ceramic coating is impregnated with catalytically active material. The ceramic coating and thus the catalytically active material are present on every wall within the structured catalyst **10** over which the gas flow flows during operation and interacts with the heated surface of the structured catalyst and the catalytically active material.

Thus, during use in a reactor system for steam reforming, a hydrocarbon feed gas flows through the channels **70** and interacts with the heated surface of the structured catalyst and with the catalytically active material supported by the ceramic coating.

In the structured catalyst **10** shown in FIG. 6 a slit **60** has been cut into the structured catalyst **10**. This slit **60** forces a current to take a zigzag route, in this instance downwards and subsequently upwards, within the macroscopic structure thereby increasing the current path and thus the resistance and consequently the heat dissipated within the macroscopic structure. The slit **60** within the macroscopic structure may be provided with embedded insulating material in order to ensure that no current flows in the transverse direction of the slit **60**.

The channels **70** in the structured catalyst **10** are open in both ends. In use of the structured catalyst in a reactor system, a hydrocarbon feed gas flows through the unit, in the direction shown by arrows **11** and **12** in FIGS. 1a and 1b, and gets heated via contact with the walls **75** of the channels **70** and by heat radiation. The heat initiates the desired steam reforming process. The walls **75** of the channels **70** may e.g. have a thickness of 0.5 mm, and the ceramic coating coated onto the walls **75** may e.g. have a thickness of 0.1 mm. Even though the arrows **11** and **12** (see FIGS. 1a and 1b) indicate that the flow of the hydrocarbon feed gas is down-flow, the opposite flow direction, viz. an up-flow, is also conceivable.

FIG. 7 shows the structured catalyst **10** of FIGS. 1a and 1b in a perspective view and with connectors **7** attached. The connectors **7** each connect a part of the structured catalyst **10** to a conductor **40**. The conductors **40** are both connected to a power supply (not shown). Each of the connectors **7** are connected to an upper part of the structured catalyst. When the conductors **40** are connected to a power supply, an electrical current is led to the corresponding connector **7** via the conductor and runs through the structured catalyst **10**. The slit **60** hinders the current flow in a transverse direction (horizontal direction of FIG. 7) throughout its lengths along the height *h* of the structured catalyst **10**. Therefore, the current runs in a direction downwards as seen in FIG. 7 in the part of the structured catalyst along the slit **60**, subsequently it runs transversely to the longitudinal direction below the slit **60** as seen in FIG. 7 and finally the current runs upwards in the longitudinal direction of the structured catalyst to the other connector **7**. The connectors **7** in FIG. 7 are mechanically fastened to the structured catalyst by means of inter alia mechanical fastening means such as screws and bolts. However, additional or alternative fastening means are conceivable. In an embodiment, the electrical power supply generates a voltage of 3V and a current of 400 A. The connectors **7** are e.g. made in materials like iron, aluminum, nickel, copper or alloys thereof.

As mentioned, the structured catalyst **10** is coated with a ceramic coating, such as an oxide, supporting the catalytically active material. However, the parts of the structured catalyst **10**, which are connected to the connectors **7**, should not be coated with an oxide. Instead, the macroscopic

24

structure of the structured catalyst should be exposed or connected directly to the connectors **7** in order to obtain a good electrical connection between the macroscopic structure and the connector.

When the connectors **7** and thus the conductors **40** are connected to the same end of the structured catalyst **10**, viz. the upper end as seen in FIG. 7, the gas entering into a reactor system housing the structured catalyst **10** would be able to cool the connectors **7** and the conductors **40**. For instance, the hydrocarbon gas entering into such a reactor system would have a temperature of 400° C. or 500° C. and would thus keep the connectors **7** and conductors **40** from reaching temperatures much higher than this temperature.

FIG. 8 shows another embodiment of a structured catalyst **10'''** with connectors **7'''**. The structured catalyst **10'''** is e.g. the structured catalyst as shown in FIG. 6. Each of the connectors **7'''** has three holes at an upper side thereof for connection to conductors (not shown). A piece of electrically insulating material **61** is inside the slit **60** (see FIG. 6) of the structured catalyst **10'''**.

FIG. 9 shows an example of the current invention as generated from experimental data on a laboratory embodiment of the invention. In a first state A the system is operating at a fixed feed flow of 100 NI/h having a composition of 31.8% CH₄, 2.4% H₂, and 66.8% H₂O at a total pressure of 10 barg and an inlet temperature of 200° C.

In this stage a current of 81 A runs in the monolith at a voltage drop of 1V. At the time mark of 10 min, the electrical potential is increased to 2.6 V over a few seconds, which gives a new current of 218 A defining stage B.

This results in an immediate increase in temperature and after 20 min this has increased from 350° C. to 875° C. and is within 10% of the steady state process value of the given condition set at 950° C. The final adjustment towards the steady state value is a function of thermal equilibration of the system and a complete heating to steady state of all elements in the system, including insulation and reactor elements. The open diamond symbols of the figure additionally shows the so called equilibrium temperature of the steam reforming reaction as evaluated by GC-TCD during the run, which is found to match the operating temperature closely. This means that the product gas of the system constantly is close to equilibrium, even during the change of operating stages. When observed, the off-set between the equilibrium temperature and operating temperature is only an experimental artefact of a delay in gas sampling relative to the temperature sampling.

FIG. 10 shows an example of the current invention as generated from experimental data on a laboratory embodiment of the invention. In a first state A the system is operating at a fixed feed flow of 100 NI/h having a composition of 31.8% CH₄, 2.4% H₂, and 66.8% H₂O at a total pressure of 2.5 barg and an inlet temperature of 200° C.

In this stage a current of 20 A runs in the monolith at a voltage drop of 2.7 V. At the time mark of 16 min, the electrical potential is decreased to 1.8 V, which gives a new current of 140 A defining stage B.

This results in an immediate decrease in temperature and after 30 min this has decreased from 950° C. to 650° C. and is within 5% of the steady state process value of the given condition set at 630° C. The final adjustment towards the steady state value is a function of thermal equilibration of the system and a complete cooling to steady state of all elements in the system, including insulation and reactor elements. The open diamond symbols of the figure additionally shows the so called equilibrium temperature of the steam reforming reaction as evaluated by GC-TCD during

25

the run, which is found to match the operating temperature closely. This means that the product gas of the system constantly is close to equilibrium, even during the change of operating stages.

The present invention has been described with reference to a number of aspects and embodiments. These aspects and embodiments may be combined at will by the person skilled in the art while remaining within the scope of the patent claims.

The invention claimed is:

1. A method for rapidly switching a metal-catalysed steam methane reforming reaction of a feed gas comprising hydrocarbons in a reactor system from a first steady-state reaction condition (A) to a second steady-state reaction condition (B) or vice-versa;

wherein said reactor system comprises a pressure shell housing a structured catalyst arranged to catalyze steam reforming of a feed gas comprising hydrocarbons, said structured catalyst comprising a macroscopic structure of an electrically conductive material, said macroscopic structure supporting a ceramic coating, where said ceramic coating supports a catalytically active material and wherein said reactor system is provided with heat insulation between said structured catalyst and said pressure shell, and where a power supply placed outside said pressure shell is arranged to supply electrical power via electrical conductors connecting to said structured catalyst, allowing an electrical current to run through said macroscopic structure, thereby heating at least part of the structured catalyst;

said method comprising the steps of:

in said first steady-state reaction condition (A):

supplying said feed gas to the reactor system in a first total flow, and

supplying a first electrical power via electrical conductors connecting an electrical power supply placed outside said pressure shell to said structured catalyst, thereby allowing a first electrical current to run through said electrically conductive material,

thereby heating at least part of the structured catalyst to a first temperature at which said feed gas is converted to a first product gas mixture over said structured catalyst under said first steady-state reaction conditions (A); and said first product gas is outlet from the reactor system;

and, in said second steady-state reaction condition (B):

supplying said feed gas to the reactor system in a second total flow,

supplying a second electrical power via electrical conductors connecting an electrical power supply placed outside said pressure shell to said structured catalyst, thereby allowing a second electrical current to run through said electrically conductive material,

thereby heating at least part of the structured catalyst to a second temperature; at which said feed gas is converted to a second product gas mixture over said structured catalyst under said second steady-state reaction conditions (B); and said second product gas is outlet from the reactor system;

wherein said second electrical power is higher than said first electrical power; and/or said second total flow is higher than said first total flow.

26

2. The method according to claim 1, wherein said at least two conductors are connected to the structured catalyst at a position on the structured catalyst closer to said first end of said structured catalyst than to said second end of said structured catalyst, and wherein the structured catalyst is constructed to direct an electrical current to run from one conductor substantially to the second end of the structured catalyst and return to a second of said at least two conductors.

3. The method according to claim 1, wherein the ratio of total gas feed flow in said first reaction condition A to said second reaction condition B (A:B) is at least 1:10.

4. The method according to claim 1, wherein the product gas outlet temperature from the structured catalyst in reaction condition B is between 50° C. to 800° C. higher than the product gas outlet temperature from the structured catalyst in reaction condition A.

5. The method according to claim 1, wherein the switch between reaction condition A and B includes a gradual change of the total gas feed flow from said first total flow to said second total flow and simultaneous gradual change of the applied electrical potential over said electrically conductive material from said first to said second electrical power.

6. The method according to claim 1, wherein the product gas outlet temperature from the structured catalyst in reaction condition B is no more than 50° C. higher than the product gas outlet temperature from the structured catalyst in reaction condition A.

7. The method according to claim 1, wherein a proportional-integral-derivative (PID) controller controls the electrical potential based on feedback reading of the process value of product gas outlet temperature from the structured catalyst.

8. The method according to claim 1, wherein the product gas outlet temperature from the structured catalyst is measured directly beneath or on the most downstream surface of the structured catalyst.

9. The method according to claim 1, wherein the switch between reaction condition A and B takes place over a period of less than 3 hours.

10. The method according to claim 1, wherein the switch between reaction condition A and B involves supplying a second electrical power to the structured catalyst.

11. The method according to claim 1, wherein the switch between reaction condition A and B comprises a transition state between said reaction conditions A and B; said transition state comprising a first period in which the electrical power is switched off, followed by a second period in which said second electrical power of condition B is supplied to the structured catalyst.

12. The method according to claim 1, wherein the switch between reaction condition A and B comprises a transition state between said reaction conditions A and B; said transition state comprising a first period in which a third electrical power is supplied to the structured catalyst, followed by a second period in which said second electrical power of condition B is supplied to the structured catalyst, said third electrical power being higher than the second electrical power.

* * * * *